

NEUR 411 · NEUROETHOLOGY · FALL 2026

Lecture 17

Bat echolocation II: FM bats, target-range computation, clutter rejection

Friday, October 2, 2026

Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering



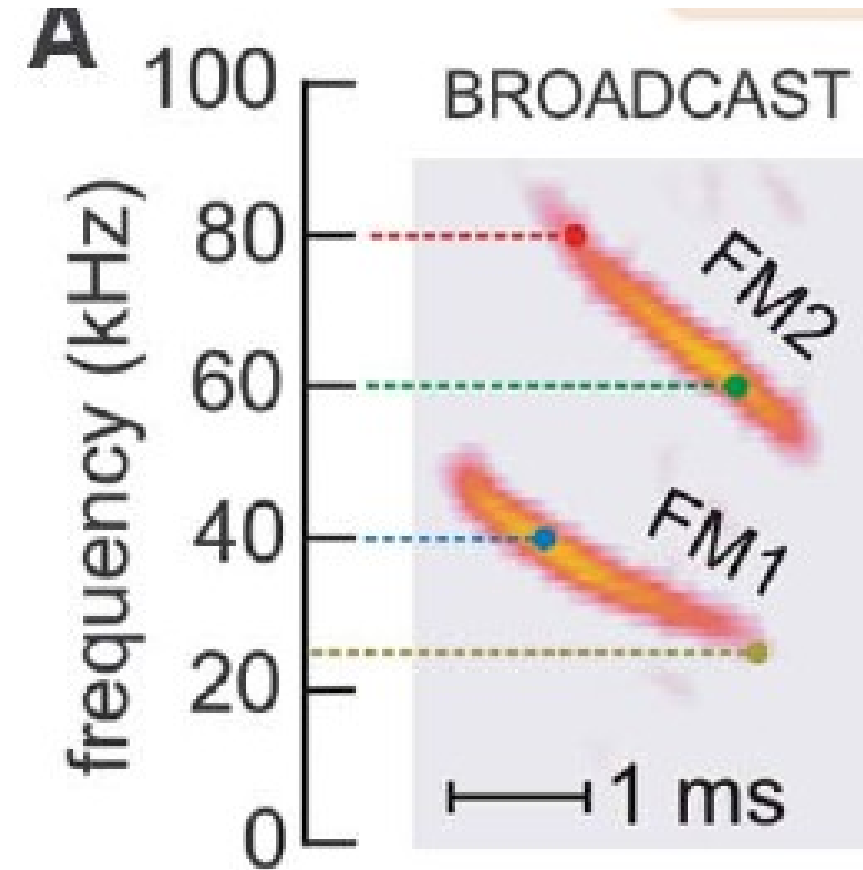
Photo: Big brown bat (*Eptesicus fuscus*).

FM calls distribute acoustic energy across frequencies

Echolocation is active sensing with emitted sounds and returning echoes. A **frequency-modulated sweep**, or FM sweep, changes frequency during a call. Big brown bats, *Eptesicus fuscus*, emit downward sweeps with two prominent **harmonics**, components related by a twofold frequency ratio.

Ming and colleagues describe FM1 sweeping approximately from 55 to 25 kHz and FM2 from 110 to 50 kHz. These wide frequency ranges differ from a sustained constant-frequency carrier. An echo preserves information about when different frequencies were transmitted and returned.

Behavioral filtering experiments distinguish the contributions of these harmonics. FM1 can support echo-delay perception without FM2; FM2 cannot simply replace missing corresponding FM1 frequencies. The two acoustic components therefore contribute asymmetrically to the tested range judgments.



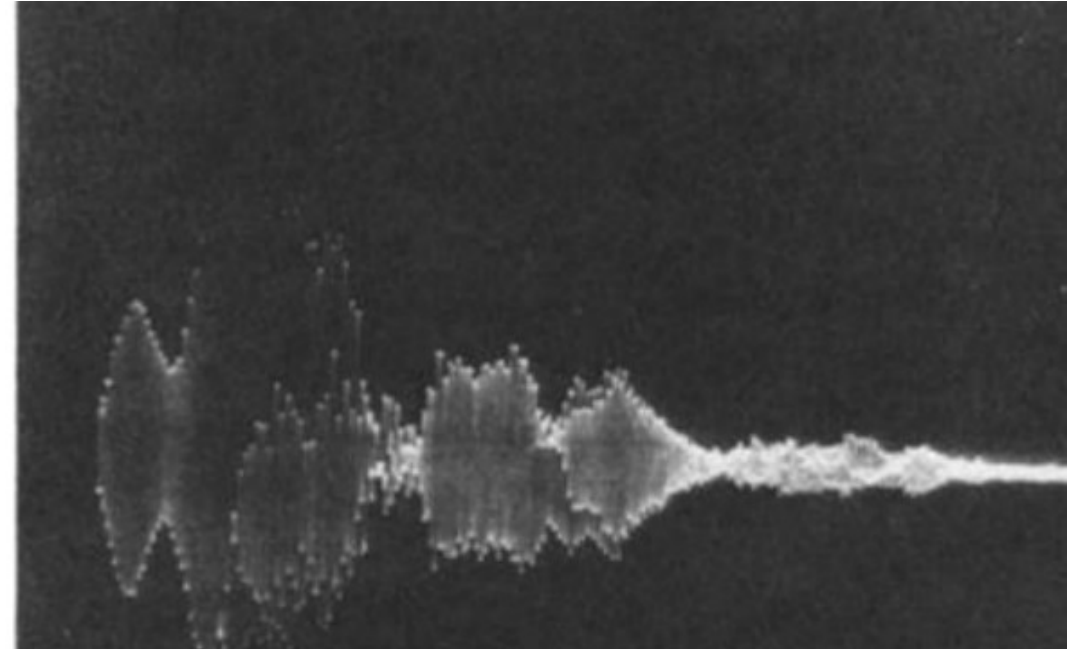
Bates et al. (2011), Fig. 1(A). Two downward FM harmonics in a big brown bat broadcast.

Echo delay specifies the round trip to a target

Echo delay is the interval between an outgoing call and its returning reflection. A more distant target produces a later echo because sound travels outward to the target and back to the bat. **Target range** is the distance from the bat to that reflector.

Simmons recorded outgoing sounds and echoes while bats discriminated target distance. A target at 30 cm returned an echo after about 1.7 ms in the experimental conditions. Electronic delay channels could change this interval while keeping the loudspeakers at fixed physical positions.

The resulting separation of acoustic delay from physical location permitted a direct behavioral test of the ranging cue. The bat could judge relative simulated distance from echo arrival time without receiving a corresponding change in the loudspeaker's actual distance.



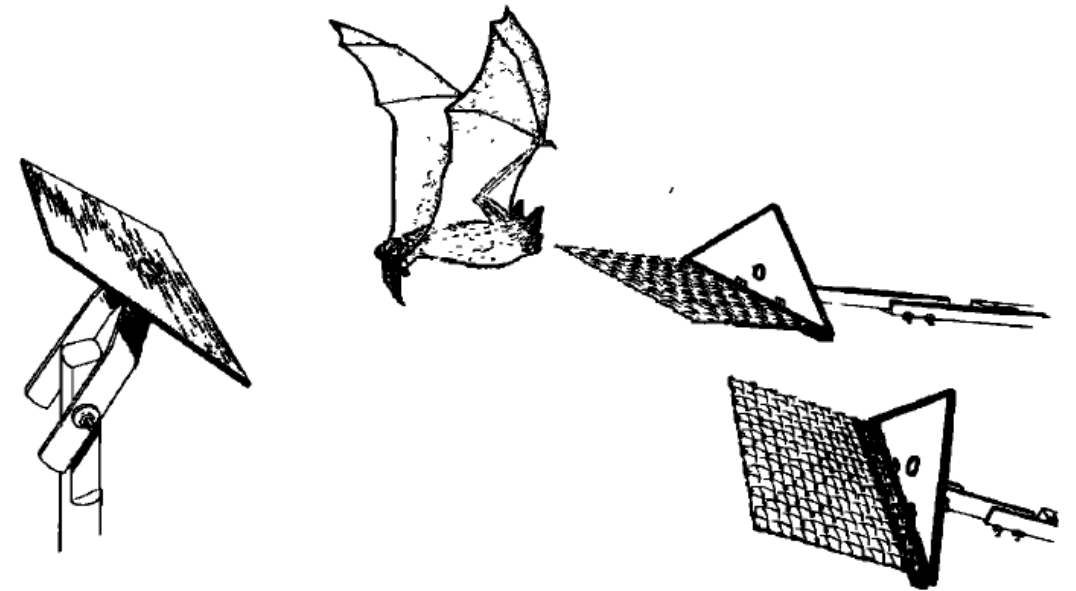
Simmons (1973), Fig. 11. Recorded outgoing sound and target echo at a range of 30 cm.

Bats can choose the nearer of two reflecting targets

Simmons trained bats to discriminate between two targets that differed in distance from a starting platform. The animal received food for moving to the landing platform associated with the nearer target. **Discrimination** here means choosing between simultaneously available alternatives.

One target remained at a fixed range, such as 30 or 60 cm, while the other was brought progressively closer to that range. Vision was excluded so that visual distance cues could not explain the choices. The triangular targets had the same form.

Performance deteriorated as the distance difference approached zero, reaching chance when both targets were equally distant. This procedure measured resolution of relative range differences. It did not measure the accuracy of an isolated absolute-distance estimate without a comparison target.



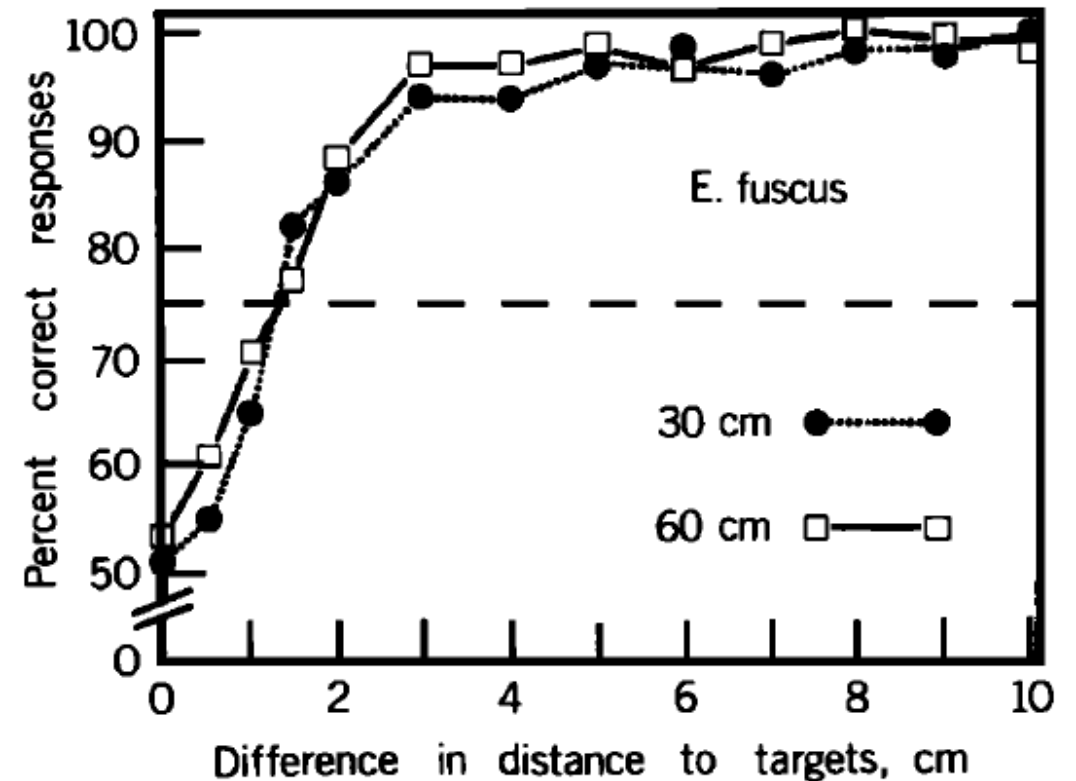
Simmons (1973), Fig. 1. Original two-target range-discrimination apparatus.

Near-range acuity remains similar at 30 and 60 cm

Range acuity is the smallest distance difference that supports a specified level of correct discrimination. Simmons used 75% correct choices as an operational criterion, between chance and nearly error-free performance in the two-target task.

Big brown bats discriminated range differences of approximately 12–14 mm at absolute ranges of 30 and 60 cm. Doubling target distance within this near-range comparison produced little change in the discrimination curve. Larger distance differences were much easier to distinguish.

The result supports approximately constant absolute resolution over these tested ranges. It does not establish that resolution remains independent of distance at every operating range. Echo audibility, propagation losses, and the conditions of the behavioral task constrain extrapolation to distant targets.



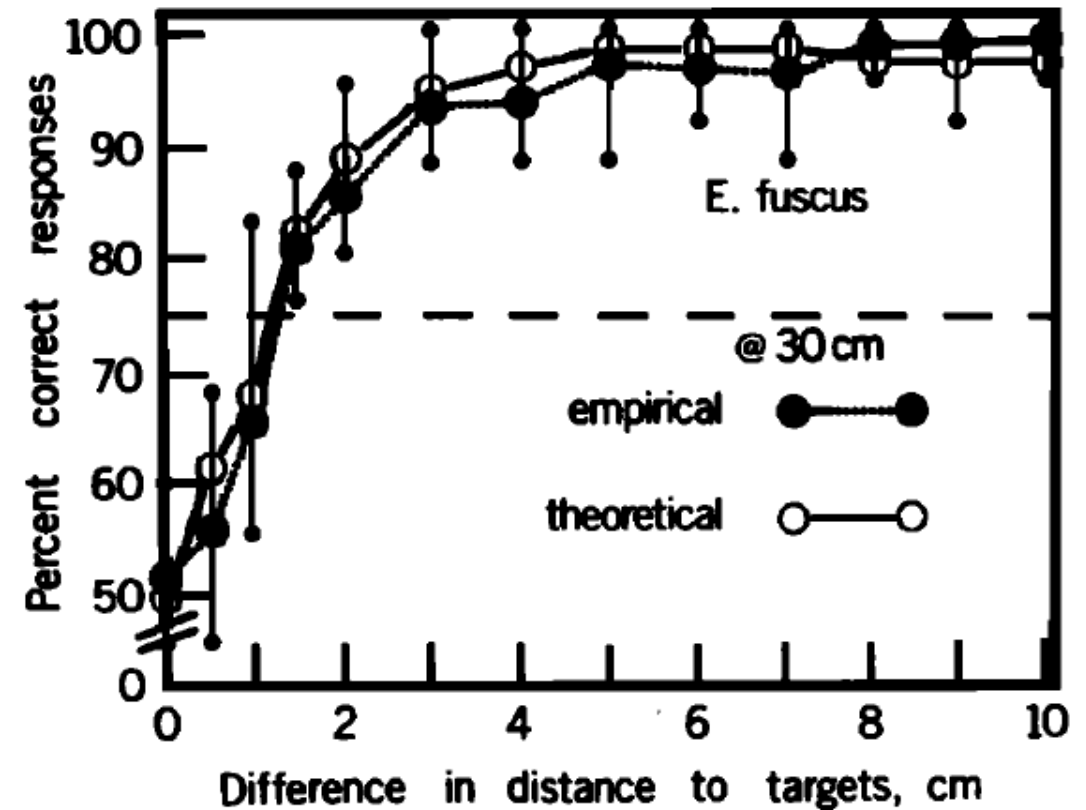
Simmons (1973), Fig. 2. Range-discrimination performance at absolute distances of 30 and 60 cm.

Behavioral resolution is narrower than target distance

At an absolute range of 30 cm, big brown bats performed almost without errors when targets differed by several centimeters. Their choices became progressively less reliable as the separation fell toward the millimeter-scale threshold. Absolute target distance and resolvable distance difference are distinct quantities.

Simmons compared this measured discrimination curve with a prediction derived from the recorded sonar signal. The prediction used the signal's **autocorrelation envelope**, a measure of similarity between a waveform and delayed versions of itself, with rapid carrier fluctuations removed.

Agreement supported a receiver matched to features of the transmitted signal. It did not directly identify the neurons performing that computation, and a correspondence between a behavioral curve and a signal-based prediction does not prove a unique cellular implementation.



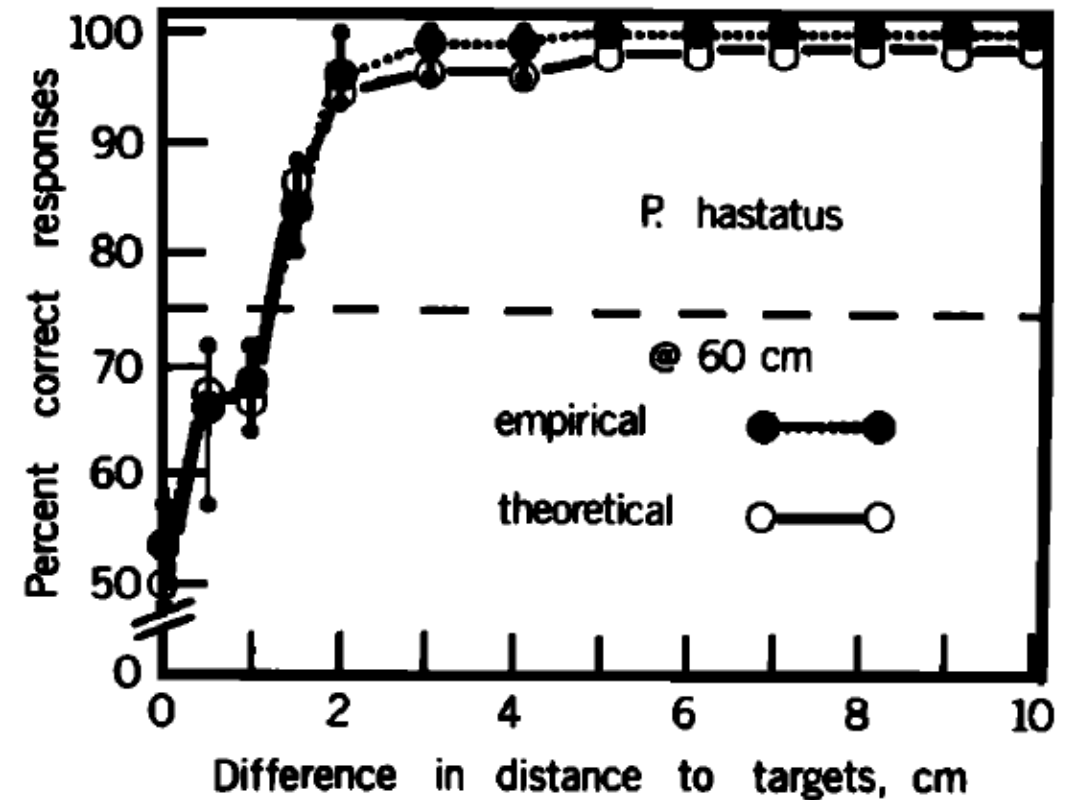
Simmons (1973), Fig. 3. Measured discrimination and the paper's signal-based prediction at 30 cm.

Range discrimination occurs across sonar signal types

Simmons compared range discrimination in bats using different sonar signals. *Eptesicus fuscus* and *Phyllostomus hastatus* used FM sounds, whereas the other tested species used combinations of constant-frequency and FM components. The species were tested with the same general nearer-target contingency.

The first three species resolved differences near 1–2 cm; *Rhinolophus ferrumequinum* had a threshold nearer 3 cm. In *Phyllostomus*, discrimination at 60 and 120 cm had similar resolution. Signal form did not divide the animals into ranging and non-ranging groups.

The paper inferred different contributions of signal components across species. These comparisons establish behavioral capabilities under the tested conditions, rather than an identical neural algorithm in all bats. Species-specific auditory processing must be distinguished from their shared ability to compare target ranges.



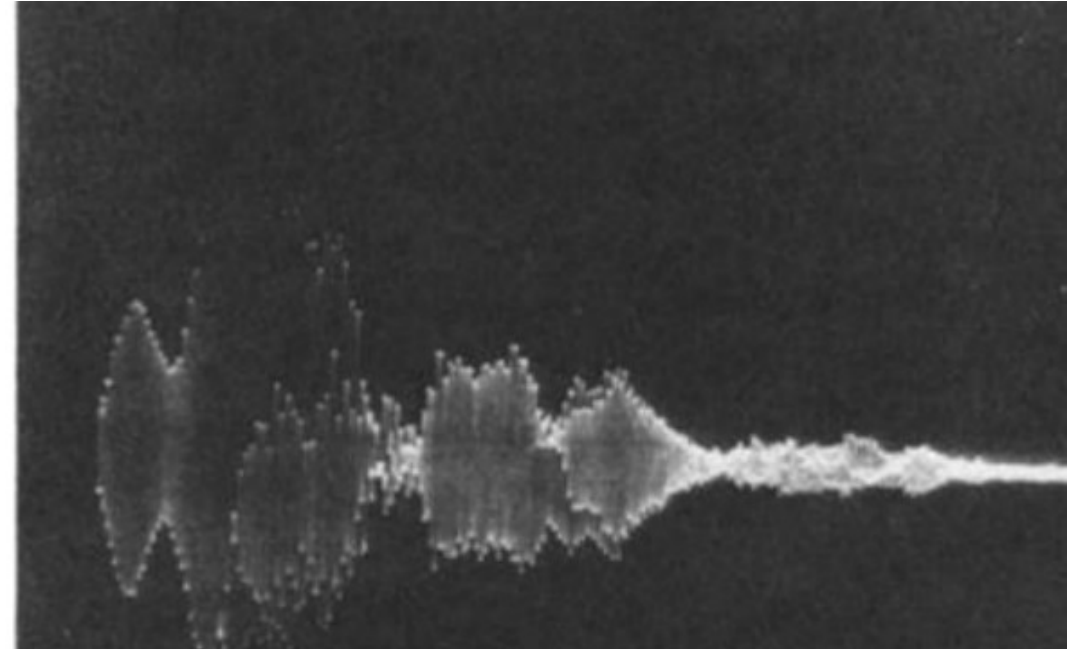
Simmons (1973), Fig. 5. Range discrimination by *Phyllostomus hastatus* at 60 cm.

Electronic echoes isolate arrival time from other cues

An **electronic target simulator** captures a bat's emitted sound, delays it, and returns it through a loudspeaker. Simmons used two independent microphone–delay–speaker channels to present echoes with different arrival times. The hardware could preserve other range-related acoustic properties while changing delay.

The loudspeakers remained fixed, and overall level and spectrum were controlled. A first arrangement with one microphone feeding both speakers disrupted the animal's ability to inspect targets separately. Using one microphone for each channel restored stable discrimination.

This control revealed that directional sampling is part of the task. A bat aims its call toward an object, changing the information returned by that object. Artificial feedback must preserve that active relationship if the experiment is to approximate separate acoustic inspection of two targets.



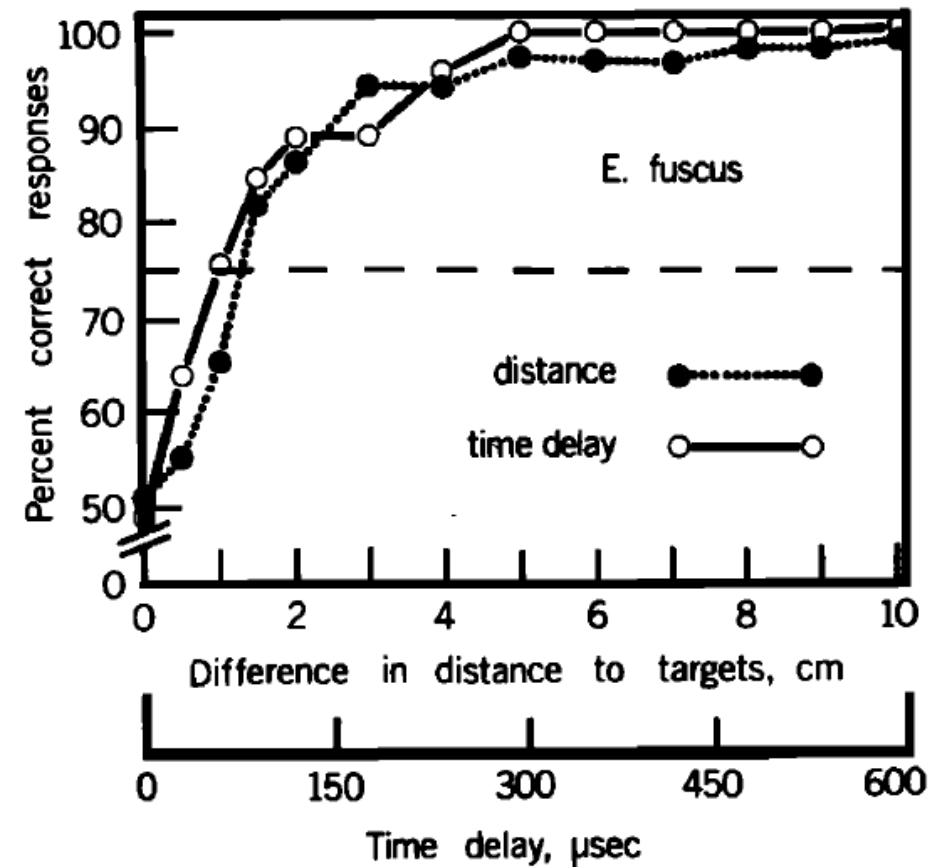
Simmons (1973), Fig. 11. Call and delayed echo recorded in the target-simulator experiment.

Echo arrival time can support the range judgment

Simmons compared physical-target discrimination with discrimination of electronically delayed echoes. In the simulator, the alternatives differed in arrival time rather than in the loudspeakers' actual positions. The animal was still rewarded for selecting the echo corresponding to the nearer target.

At a simulated range of 30 cm, the electronic-delay and physical-distance performance curves were closely similar. Arrival-time differences corresponding to about 1 cm of range could be discriminated in the controlled feedback task. The figure retains both distance and delay scales.

Echo timing is therefore sufficient to mediate this relative-range judgment in big brown bats. The experiment does not imply that bats ignore spectral, directional, or intensity information in natural scenes; those variables were controlled here to isolate a particular ranging cue.



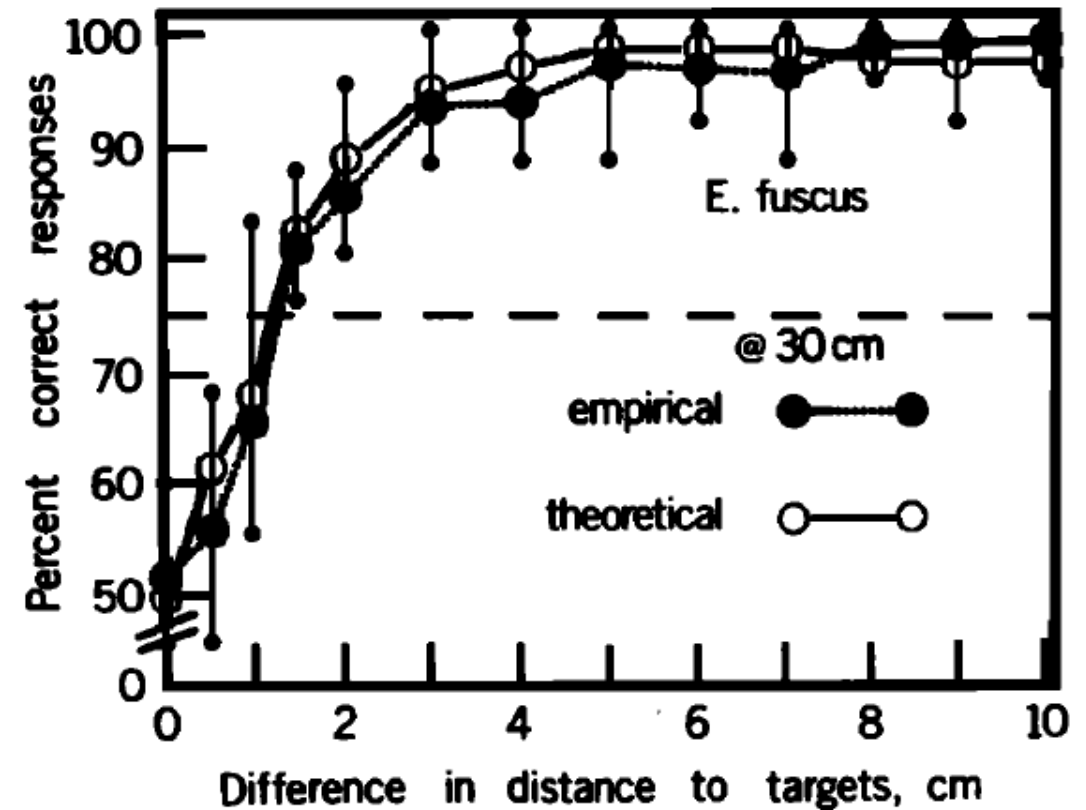
Simmons (1973), Fig. 13. Physical-range and electronic-delay discrimination at a 30-cm reference range.

A matched receiver is a functional hypothesis

A **matched receiver** processes an incoming echo in relation to the structure of the transmitted signal. **Cross-correlation** measures similarity between two signals as their relative timing is varied. Simmons proposed a neural equivalent of this operation from the relationship between acoustic structure and behavioral resolution.

The measured range-discrimination curves resembled predictions based on the envelopes of recorded calls' autocorrelation functions. The hypothesis concerns extraction of echo arrival time from a broadband signal rather than measurement of a single acoustic frequency.

The paper explicitly allowed neural representations distributed across frequency and time instead of literal waveform comparison. Behavioral agreement with a prediction supports a computational relationship, but it does not establish a permanent call template, a specific synaptic circuit, or direct neural copying of the acoustic waveform.



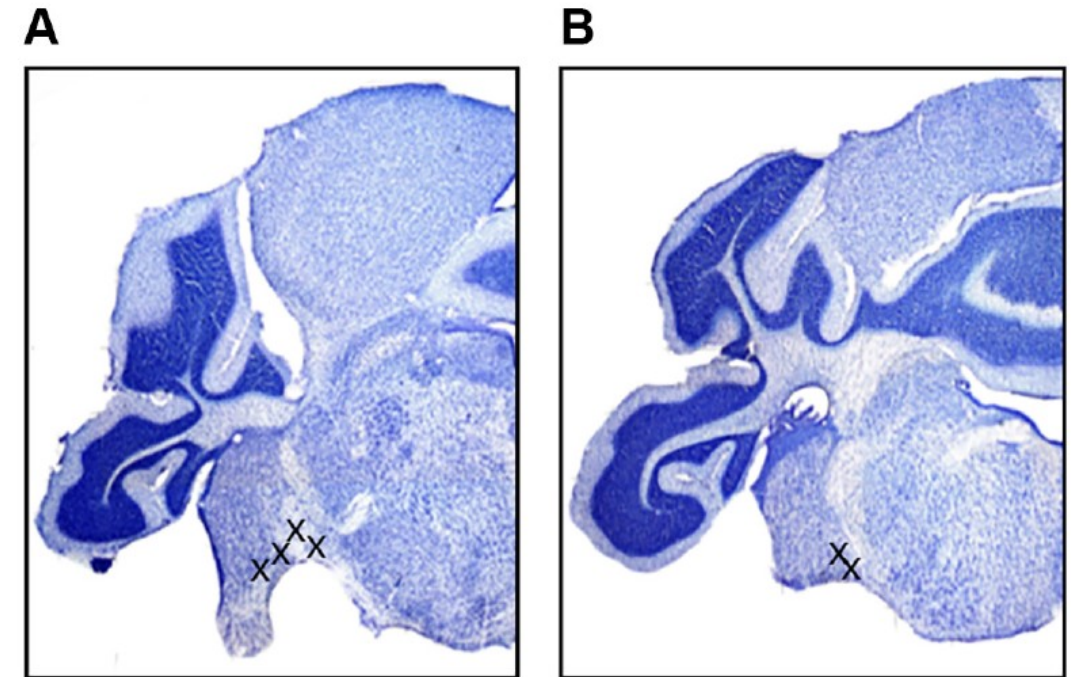
Simmons (1973), Fig. 3. Behavioral range resolution compared with the published receiver prediction.

Timing is registered early in the auditory pathway

The **cochlear nucleus** is the first central relay receiving auditory-nerve input from the ear. The **inferior colliculus** is an auditory midbrain center receiving ascending inputs from brainstem nuclei. Simmons and colleagues compared population timing in these two structures in big brown bats.

Electrodes recorded in the ventral cochlear nucleus and inferior colliculus during anesthesia. Histological lesions identified cochlear-nucleus sites near the entry of the eighth cranial nerve and near the trapezoid body. The recovered positions restricted the anatomical interpretation of the recordings.

The comparison tested whether precise timing emerged only at the midbrain or was already present in an early relay. It sampled particular regions rather than every subdivision of the cochlear nucleus, so the findings cannot be assigned uniformly to all neurons or pathways in that structure.



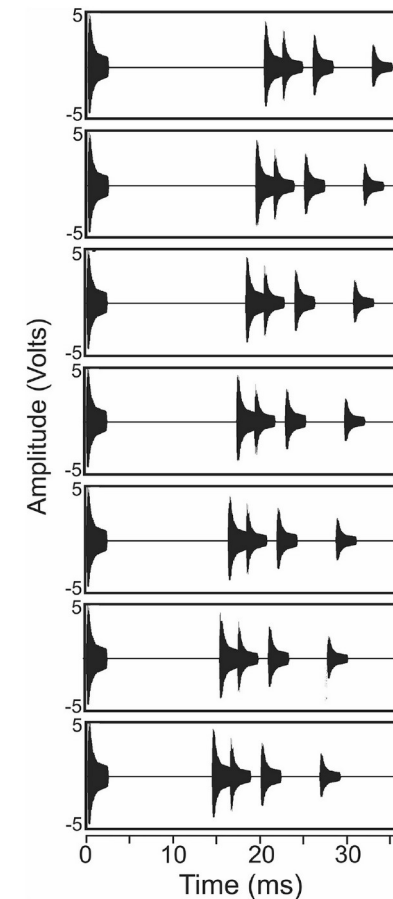
Simmons et al. (2024), Fig. 2(A–B). Histological sections locating ventral cochlear-nucleus recordings.

Echo cascades approximate a scene with several objects

An **echo cascade** is a sequence of reflections arriving after one broadcast. Simmons and colleagues constructed stimuli containing a 2.5-ms downward FM broadcast followed by four echoes with the same spectral content and different amplitudes. The stimulus approximated multiple reflectors near a flying bat.

The first echo arrived at delays from 20.4 to 14.4 ms in 1-ms steps. Subsequent echo intervals were 2.1, 3.5, and 6.8 ms. FM1 swept from 50 to 20 kHz and FM2 from 100 to 40 kHz.

Changing the first-echo delay altered the simulated scene while preserving its general cascade structure. This design tested timing during a sequence of echoes rather than an isolated tone. The anesthetized preparation did not measure ongoing vocal choice or obstacle-avoidance behavior.



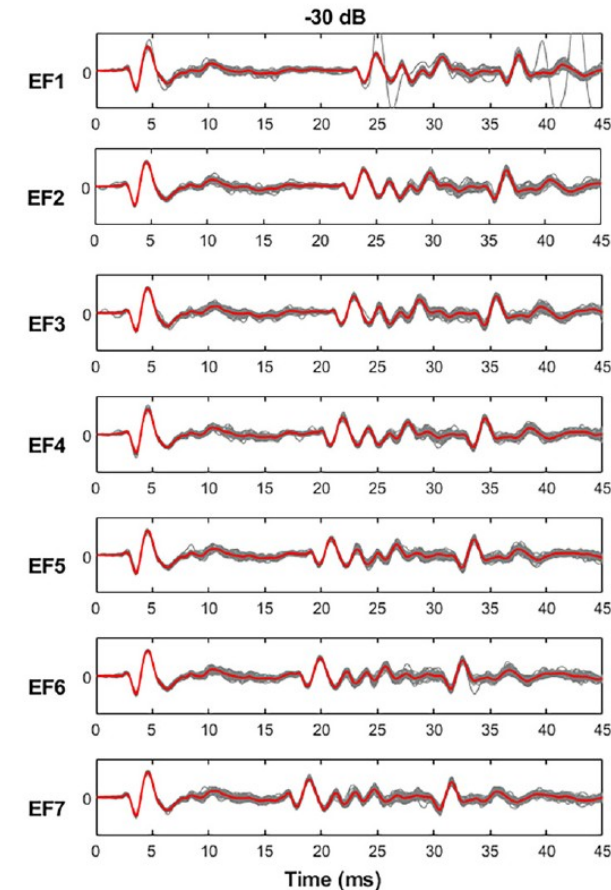
Simmons et al. (2024), Fig. 1. Published broadcast–echo waveforms with first-echo delays from 20.4 to 14.4 ms.

Population potentials register calls and echoes

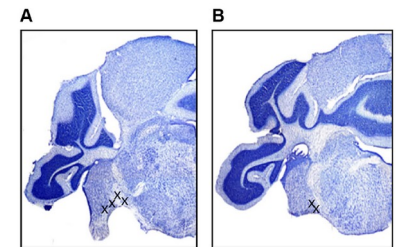
A **local field potential**, or LFP, is an extracellular voltage signal reflecting coordinated activity in a neuronal population. Simmons and colleagues filtered recordings to examine low-frequency population waveforms rather than treating every deflection as a single neuron's action potential.

In the cochlear nucleus, each broadcast and the first echo produced a prominent negative peak followed by a positive peak. Repeated presentations of the same echo-flow stimulus produced closely overlapping early responses. Later parts of the cascade had more variable waveforms.

The consistency of a population response provides a candidate timing signal for registering acoustic events. LFP timing does not identify the exact contributing cells, nor does it establish which downstream decoder reads the response. The measured signal combines neuronal activity within the electrode's recording field.



Simmons et al. (2024), Fig. 3, left column. Repeated cochlear-nucleus responses at -30 dB attenuation.



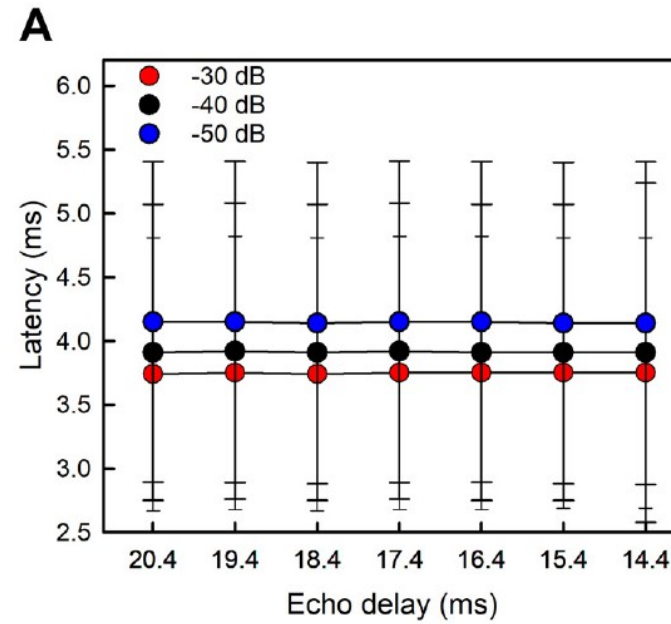
Simmons et al. (2024), Fig. 2(A–B). Histological sections locating ventral cochlear-nucleus recordings.

Broadcast latency changes with acoustic level

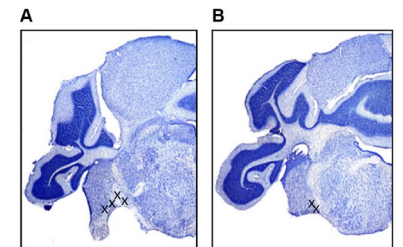
Response latency is the time from a stimulus event to a specified neural response. In the cochlear nucleus, Simmons and colleagues measured the first negative LFP peak after the broadcast. The event used for measurement was kept consistent across stimulus conditions.

Mean broadcast latencies were 3.7, 3.9, and 4.2 ms at attenuation settings of -30, -40, and -50 dB. Greater attenuation means a weaker presented sound. The broadcast's latency remained similar when only the later echo delay changed.

Acoustic level can therefore shift the timing of an early population response even when the broadcast itself starts at the same time. A timing-based range code must accommodate level-dependent response delays rather than assuming that every sound onset produces a fixed neural latency.



Simmons et al. (2024), Fig. 4(A). Cochlear-nucleus broadcast latency at three attenuation settings.



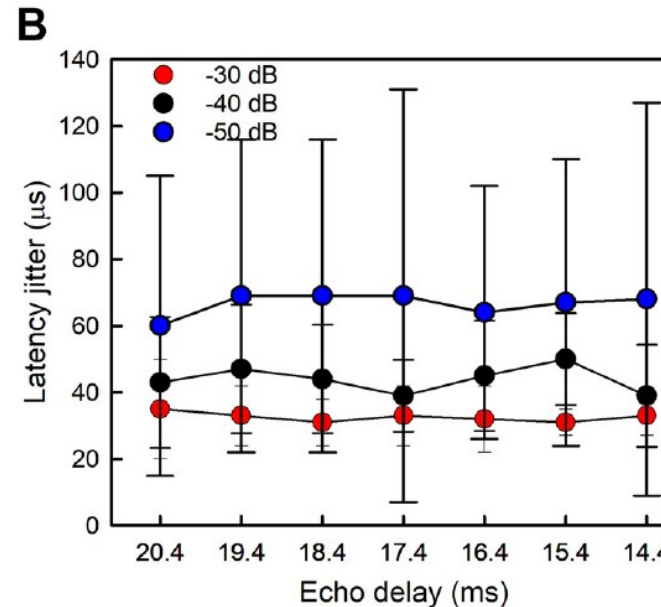
Simmons et al. (2024), Fig. 2(A–B). Histological sections locating ventral cochlear-nucleus recordings.

Broadcast timing varies by tens of microseconds

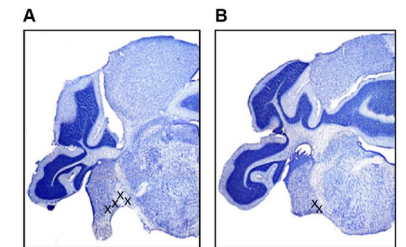
Latency jitter is the trial-to-trial spread in the timing of a defined neural response. A small jitter indicates that repeated sound presentations evoke that event at nearly the same time. It differs from the mean latency, which measures how long the response takes to arrive.

For the cochlear-nucleus broadcast peak, the paper reported mean jitter values of 32, 44, and 67 μs at -30 , -40 , and -50 dB attenuation. Timing became more variable as the stimulus weakened, while remaining comparatively stable across the seven later-echo delays.

Precision on this scale supports population timing as a potential component of echo processing. It is not the same as a behavioral threshold, and it does not demonstrate that each individual neuron in the recorded population has equally precise responses.



Simmons et al. (2024), Fig. 4(B). Cochlear-nucleus broadcast latency jitter in μs .



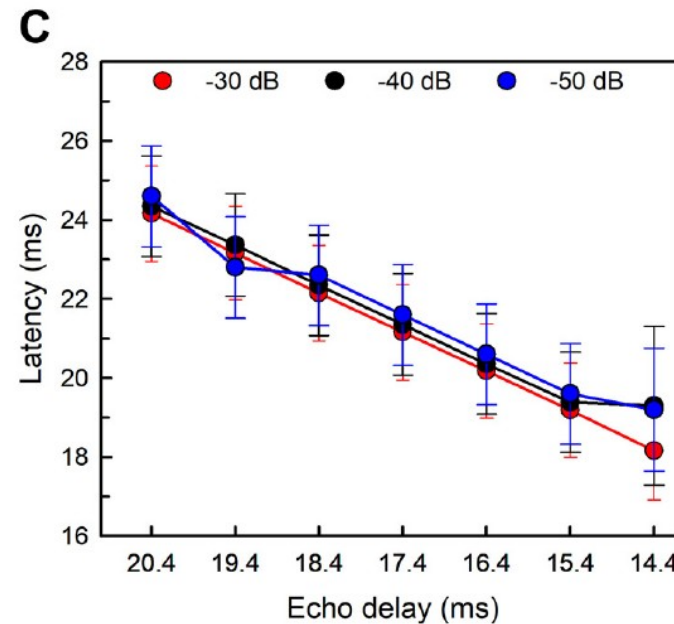
Simmons et al. (2024), Fig. 2(A–B). Histological sections locating ventral cochlear-nucleus recordings.

First-echo responses follow the changing delay

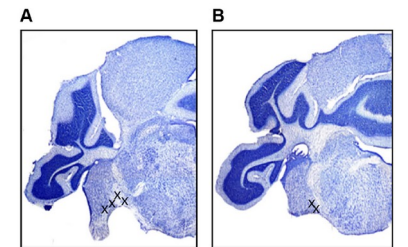
The first returning echo in the cascade moved earlier as the simulated target approached. Cochlear-nucleus response peaks consequently occurred earlier relative to the broadcast onset. The neural signal preserved the ordering of the seven experimentally imposed echo delays.

Simmons and colleagues also quantified **amplitude–latency trading**, the shift in response timing associated with a change in sound level. The first echo's cochlear-nucleus response changed by about 18 μ s per dB, while the corresponding broadcast measure was about 20 μ s per dB.

Echo timing and echo strength thus influence the measured latency together. These are separable experimental relationships, but they coexist in the auditory signal. The result supports accurate registration of relative timing while showing that response latency is not an uncorrected acoustic timestamp independent of level.



Simmons et al. (2024), Fig. 4(C). First-echo latency tracks experimentally shortened echo delay.



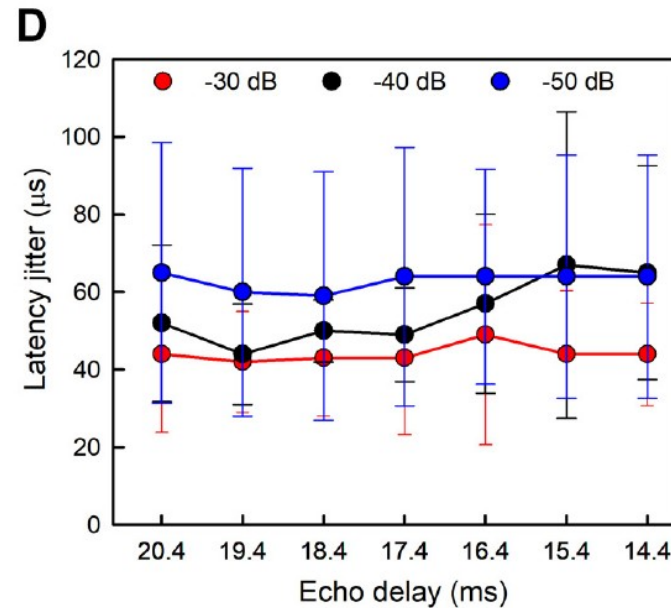
Simmons et al. (2024), Fig. 2(A–B). Histological sections locating ventral cochlear-nucleus recordings.

First-echo jitter is stable across the tested delays

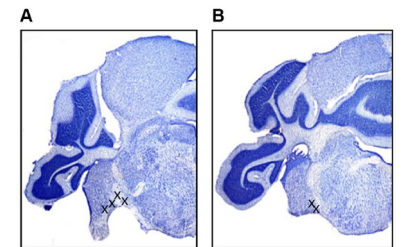
Simmons and colleagues compared trial-to-trial timing precision of the first-echo response across delays from 20.4 to 14.4 ms. The cochlear-nucleus peak shifted with acoustic delay, but its repeatability remained relatively stable across that tested range.

Mean jitter values for this peak were 44, 50, and 63 μ s at the three attenuation levels. The overall value was about 54 μ s. Weaker stimuli increased the variability, so stability across echo delays did not imply complete independence from all acoustic variables.

The result separates the representation of a moving event from the precision of that representation. It applies to a prominent population-response peak in the ventral cochlear nucleus under anesthesia. It does not identify an individual neuron tuned exclusively to a particular target distance.



Simmons et al. (2024), Fig. 4(D). First-echo latency jitter at seven delays and three sound levels.



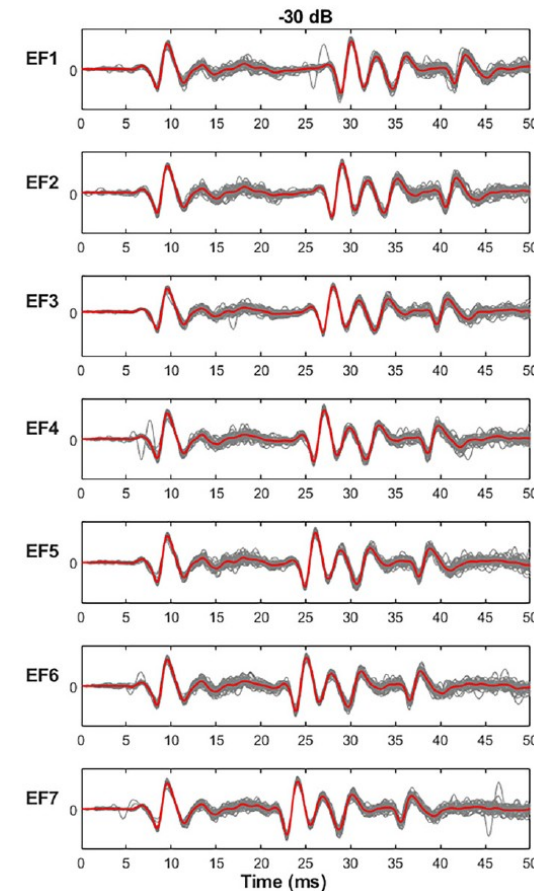
Simmons et al. (2024), Fig. 2(A–B). Histological sections locating ventral cochlear-nucleus recordings.

Midbrain responses retain precise onset timing

The inferior colliculus integrates ascending auditory information from multiple brainstem pathways. Simmons and colleagues reanalyzed midbrain LFPs from the same animals used for the cochlear-nucleus comparison, applying the same echo-flow stimuli and timing measurements.

Broadcast responses in the inferior colliculus had mean latencies of 6.7, 7.0, and 7.5 ms at the three attenuation levels. These values were longer than cochlear-nucleus latencies, consistent with additional processing before a midbrain response. Early broadcast and echo peaks remained closely aligned across repetitions.

A later mean response need not have poorer temporal precision. The study measured propagation and repeatability separately. The recordings support retention of precise population timing at the midbrain, while leaving the responsible cells and contributions of parallel ascending inputs unresolved.



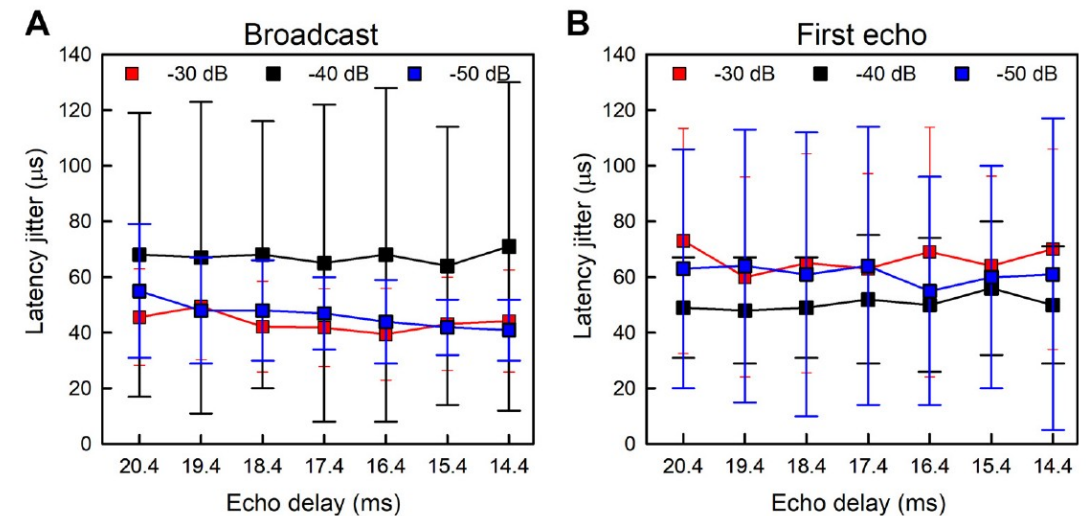
Simmons et al. (2024), Fig. 5, left column. Repeated inferior-colliculus responses at -30 dB attenuation.

Timing is similarly precise in two auditory relays

Simmons and colleagues found mean inferior-colliculus jitter values of about 52 μs for the broadcast and 59 μs for the first echo. These were similar to the corresponding cochlear-nucleus values of approximately 48 and 54 μs .

The comparison provides little evidence that timing precision was dramatically sharpened between these sampled early and midbrain populations. It is consistent with precise timing already entering the ascending pathway and being preserved. The result does not require identical response latencies at the two relays.

This inference concerns the recorded LFP events, not every neural feature in the pathway. Other regions or response measures could undergo additional transformations. A similarity between two population measurements cannot identify the anatomical projection carrying all of the precise timing information.



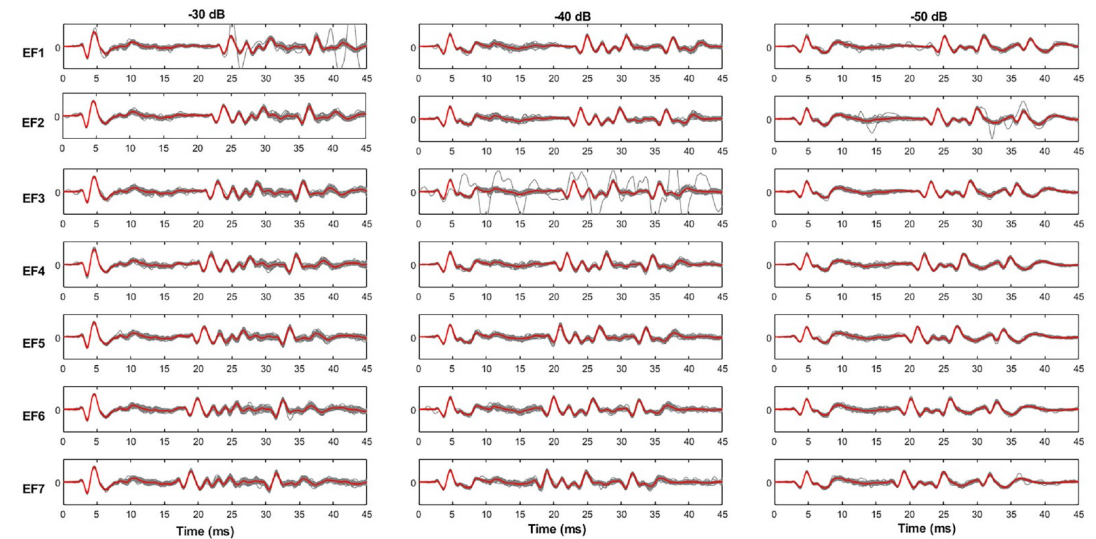
Simmons et al. (2024), Fig. 6(A–B). Midbrain latency jitter for broadcast and first-echo responses.

Later echoes have more variable population responses

A broadcast followed by several reflections recruits an evolving auditory response rather than four independent copies of one onset event. Simmons and colleagues measured prominent positive LFP peaks throughout the four-echo cascade as well as the early negative peak.

Cochlear-nucleus timing variability for the later positive peaks was around 124 μs across the cascade, and the comparable inferior-colliculus value was about 111 μs . These values were larger than the tens-of-microseconds jitter of the earliest negative peaks, while remaining below a millisecond.

The authors linked the increased variability to recruitment of additional neurons during the extended sequence. That account is an interpretation of population waveforms, rather than direct identification of every recruited cell. Precision depends on which event and which portion of a response are measured.



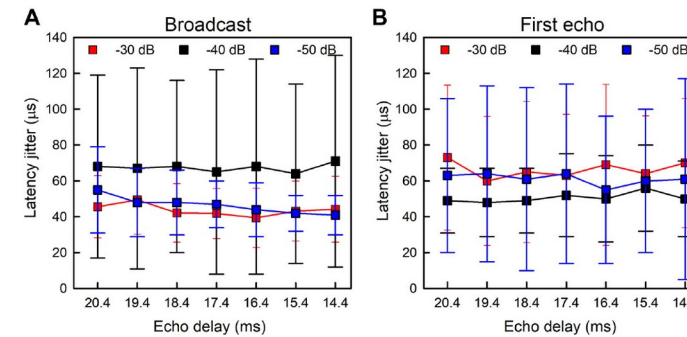
Simmons et al. (2024), Fig. 3. Cochlear-nucleus responses throughout the seven echo-flow sequences.

Measured brainstem precision leaves a perceptual gap

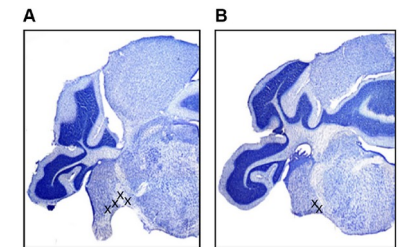
The 2024 study discusses behavioral echo-delay acuity reaching approximately 1 μ s or less in specialized psychophysical tasks. Its population-response jitter values were usually tens of microseconds. Neural repeatability approached this behavioral scale without reaching it.

The authors proposed that precisely timed population signals contribute to the eventual range computation. Combining information across frequencies and responses could improve a readout, but the experiment did not directly identify such a decoder or measure its achieved precision.

The temporal sampling and restricted recording sites also limit interpretation. A 20-kHz digitization rate places important constraints on very small timing estimates. The established finding is precise early and midbrain population activity; the complete mechanism of perceptual hyperacuity remains unresolved.



Simmons et al. (2024), Fig. 6(A–B). Midbrain latency jitter for broadcast and first-echo responses.



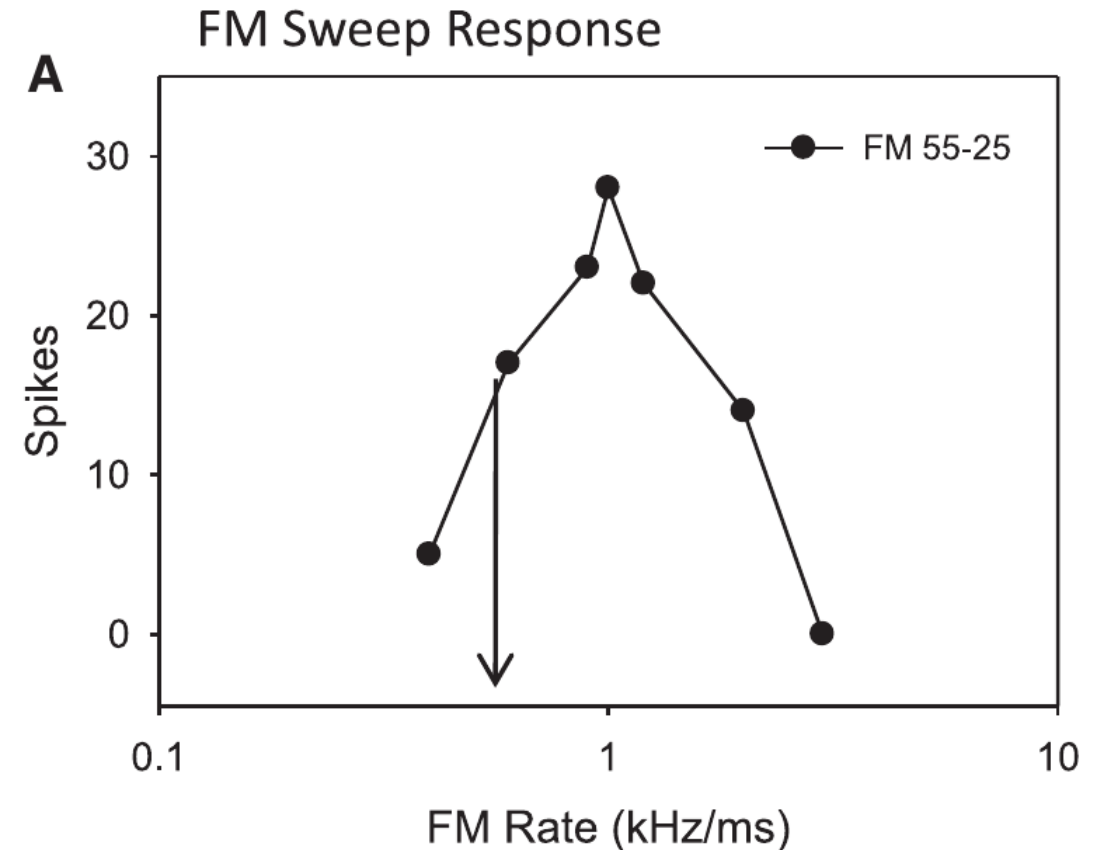
Simmons et al. (2024), Fig. 2(A–B). Histological sections locating ventral cochlear-nucleus recordings.

Pallid-bat midbrain neurons select downward sweeps

Sweep direction selectivity is a difference in neuronal response to upward and downward frequency changes. Williams and Fuzessery studied this property in the inferior colliculus of the pallid bat, *Antrozous pallidus*, whose echolocation call sweeps downward approximately from 60 to 30 kHz.

Extracellular electrodes recorded individual neurons in the high-frequency midbrain region. Broad sweeps were presented over durations from 0.5 to 100 ms, allowing direction and **sweep rate**, the frequency change per unit time, to vary. Responses were measured near the neuron's auditory threshold.

Many recorded neurons responded strongly to downward sweeps and weakly to upward sweeps. The experiment isolated inhibitory-sideband mechanisms; neurons whose selectivity depended on duration tuning or asymmetrical facilitation were excluded. Its conclusions therefore concern a defined subset of FM-sensitive neurons.



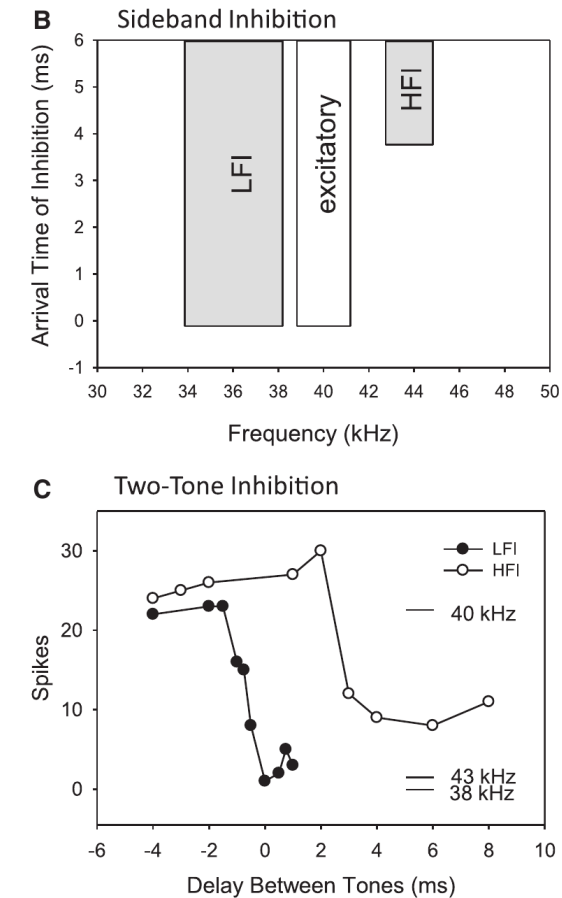
Williams & Fuzessery (2011), Fig. 2(A). A pallid-bat neuron's response depends on FM sweep rate.

Inhibitory sidebands flank the excitatory band

An **excitatory frequency band** contains sounds that evoke a neuron's spikes. An **inhibitory sideband** contains frequencies that suppress an otherwise effective response. The inhibitory frequencies need not produce a visible excitatory response when presented alone.

Williams and Fuzessery mapped sidebands on either side of a neuron's preferred frequencies using pairs of tones. **Low-frequency inhibition**, or LFI, generally arrived early relative to excitation. **High-frequency inhibition**, or HFI, generally arrived later and could suppress slow downward sweeps.

A sweep crosses these frequency regions in an ordered sequence. Reversing sweep direction changes when the inhibitory and excitatory inputs are activated. Changing sweep rate alters the time between them. Frequency tuning and relative input timing therefore interact to select particular FM sounds.



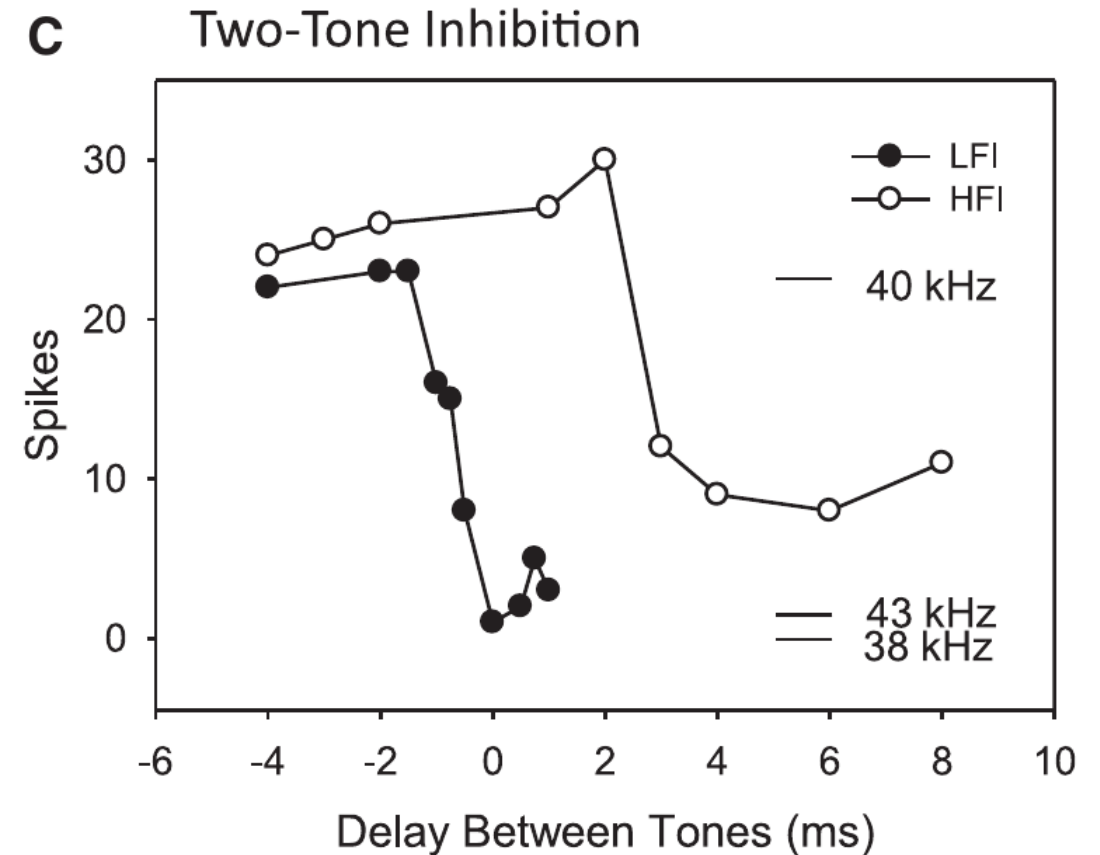
Williams & Fuzessery (2011), Fig. 2(B–C). Frequency-sideband organization and two-tone suppression timing.

Two-tone tests measure relative inhibitory arrival

Williams and Fuzessery paired a 5-ms excitatory tone with a 20-ms tone in an inhibitory sideband. They varied the interval between the tones and measured how much the neuron's response to the excitatory sound was reduced.

Suppression despite a later inhibitory-tone onset indicates that the inhibitory pathway can act sooner than the excitatory pathway. If the inhibitory sound must begin earlier, its effect arrives relatively late. The measured quantity is a functional timing relationship, not a direct recording of a synaptic current.

This **two-tone inhibition** method predicted which frequency sequences would be suppressed. A low-frequency tone with early inhibitory action can block an upward sweep before it reaches excitation. A high-frequency tone with delayed action can preferentially block a slow downward sweep.



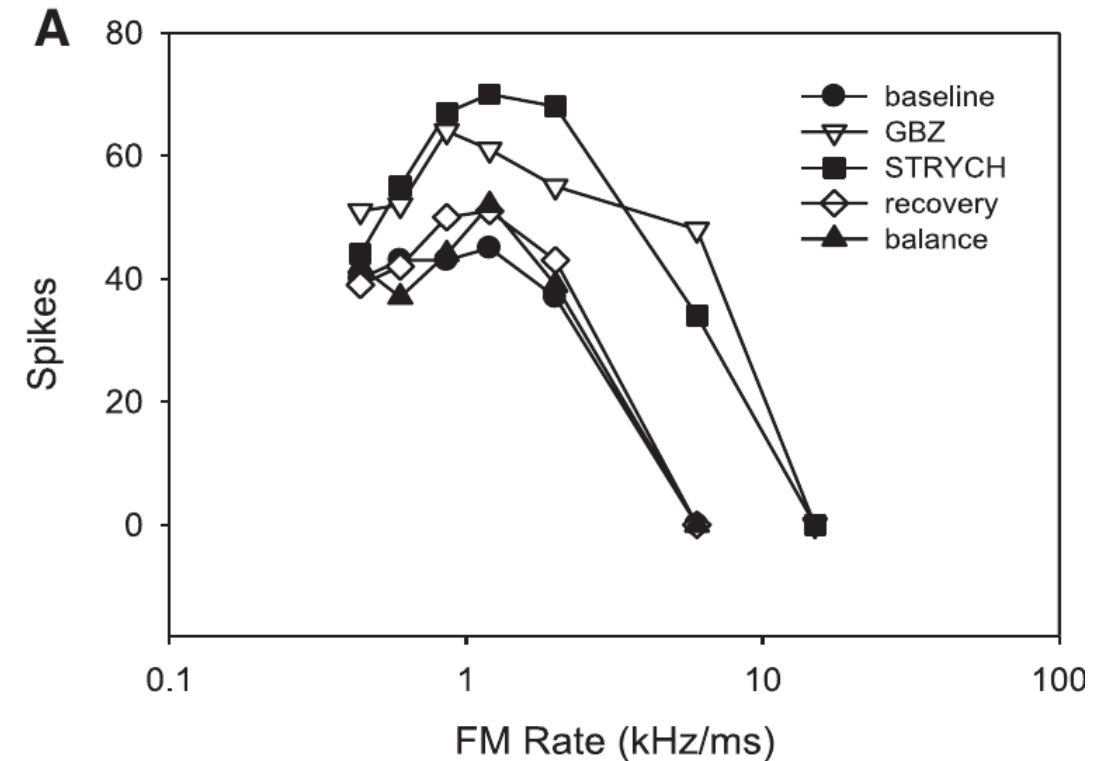
Williams & Fuzessery (2011), Fig. 2(C). Suppression varies with the interval between excitatory and inhibitory tones.

Receptor blockers test local inhibitory inputs

GABA and **glycine** are inhibitory neurotransmitters in the auditory pathways studied here. Their ionotropic GABA_A and glycine receptors mediate inhibitory membrane conductances. **Gabazine** blocks GABA_A receptors; **strychnine** blocks glycine receptors, permitting tests of each input's contribution.

Williams and Fuzessery delivered the blockers near recorded neurons by **microiontophoresis**, movement of drug from a small pipette using electrical current. They compared baseline, drug application, and recovery. A balance-electrode current control tested whether current alone changed the response.

Responses commonly increased during receptor blockade and returned toward baseline after application stopped, typically over 10–30 minutes. These controls support a receptor-dependent local effect. Increased firing alone does not establish loss of selectivity; responses to different sweep directions and rates must also be compared.



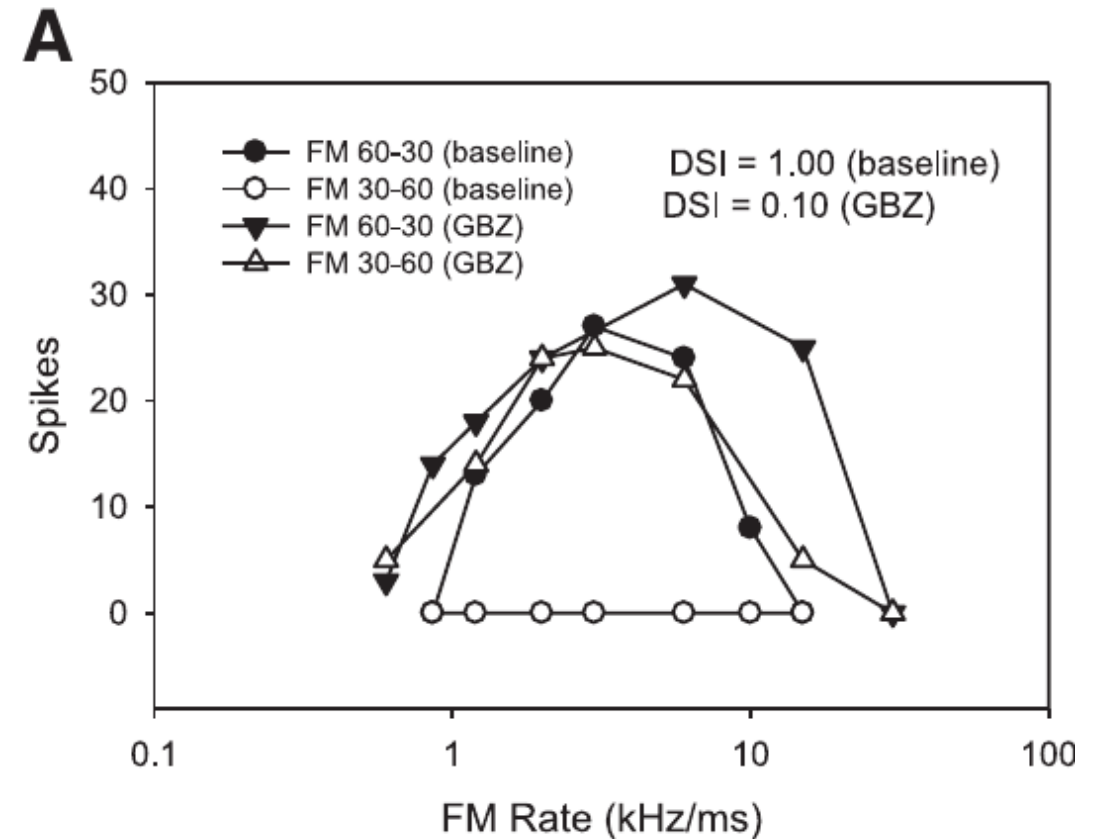
Williams & Fuzessery (2011), Fig. 1(A). Responses during baseline, receptor blockade, recovery, and current control.

Blocking inhibition can admit upward FM responses

Downward-selective neurons normally responded weakly to upward sweeps even when the sweeps crossed the excitatory band. After local GABA_A or glycine receptor blockade, many neurons produced stronger upward-sweep responses and lost much of their direction selectivity.

Williams and Fuzessery compared both directions across sweep rates, rather than inferring selectivity from the response to one sound. Gabazine and strychnine could each reduce the preference for downward sweeps. Combined application removed the preference in an even larger fraction of the tested neurons.

The appearance of responses to previously ineffective upward sweeps supports a role for local inhibition in creating or refining direction selectivity. The result does not imply that every direction-selective neuron depends on both receptor classes or that all FM selectivity originates in the inferior colliculus.



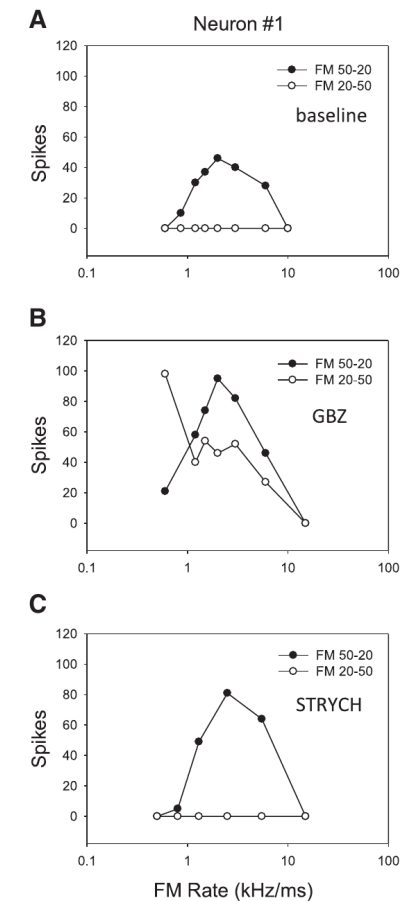
Williams & Fuzessery (2011), Fig. 3(A). Gabazine alters responses to upward and downward FM sweeps.

Neurons depend on different inhibitory inputs

Williams and Fuzessery applied gabazine and strychnine separately to the same neurons. This within-neuron comparison tested whether an apparent population similarity concealed differences in the receptor pathways supporting individual neurons' preferences.

In one published example, gabazine removed direction selectivity whereas strychnine did not. Baseline direction preference was therefore compatible with several inhibitory-input arrangements rather than one uniform transmitter mechanism.

The measurements identify functional dependence on receptor classes at the recording site. They do not identify every presynaptic source or establish that receptor distribution alone determines selectivity. Spectral position and relative timing of the inhibitory input also influence whether that input can suppress a sweep.



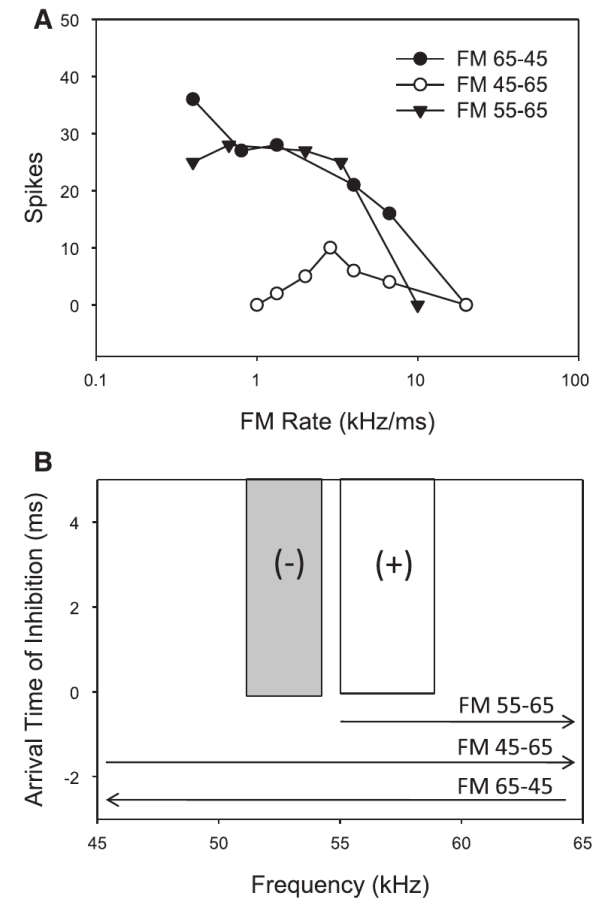
Williams & Fuzessery (2011), Fig. 4(A–C). One neuron loses direction selectivity with gabazine but retains it with strychnine.

Removing low frequencies removes direction tuning

An upward sweep encounters low frequencies before reaching a neuron's excitatory frequency band. If low-frequency inhibition acts early, it can prevent spikes when excitation arrives. A downward sweep reaches excitation before crossing that low-frequency inhibitory region.

Williams and Fuzessery tested this relationship by removing low frequencies from upward sweeps. A published neuron responded similarly to upward and downward sweeps when frequencies below 55 kHz were excluded from the upward stimulus. Those excluded frequencies overlapped its low-frequency inhibitory sideband.

Changing the stimulus spectrum thus removed a timing-dependent source of suppression without using a drug. The finding supports the sideband mechanism, but by itself cannot locate where the inhibitory selectivity is constructed. Local pharmacological tests are needed to assess the inferior-colliculus contribution.



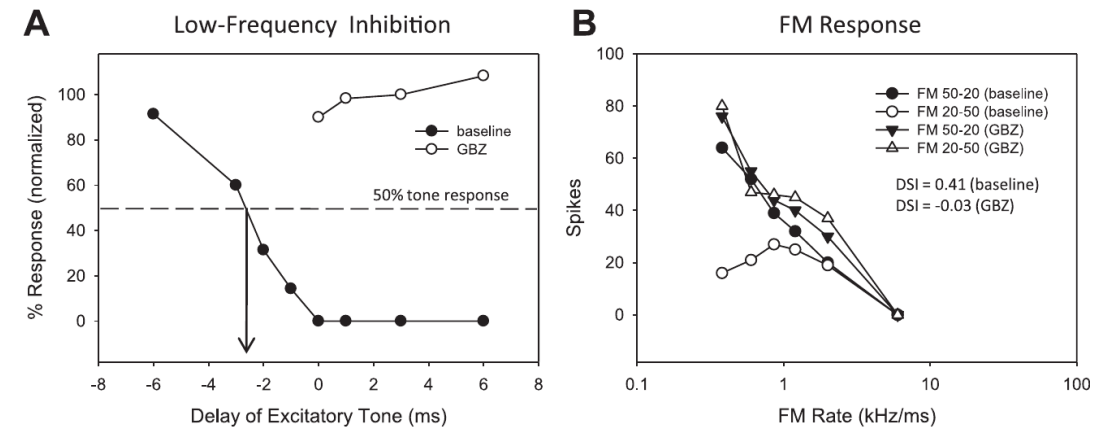
Williams & Fuzessery (2011), Fig. 5(A-B). Removing the low-frequency inhibitory region permits upward-sweep responses.

Low-frequency inhibition shapes direction tuning

Williams and Fuzessery measured low-frequency two-tone suppression before and during receptor blockade, then tested the same neuron's responses to upward and downward sweeps. This linked a change in the inferred inhibitory mechanism to a change in FM selectivity.

When gabazine eliminated low-frequency inhibition in the published example, an inhibitory tone no longer suppressed excitation. The neuron simultaneously gained upward-sweep responses and lost its strong direction preference. Similar associations occurred across the relevant neurons tested with either blocker.

The paired measurements support LFI as a causal contributor to direction selectivity in these neurons. They are more specific than a general increase in firing after a drug. The experiment still does not reveal every cellular source of the inhibitory input or exclude other mechanisms in neurons outside the sampled subset.



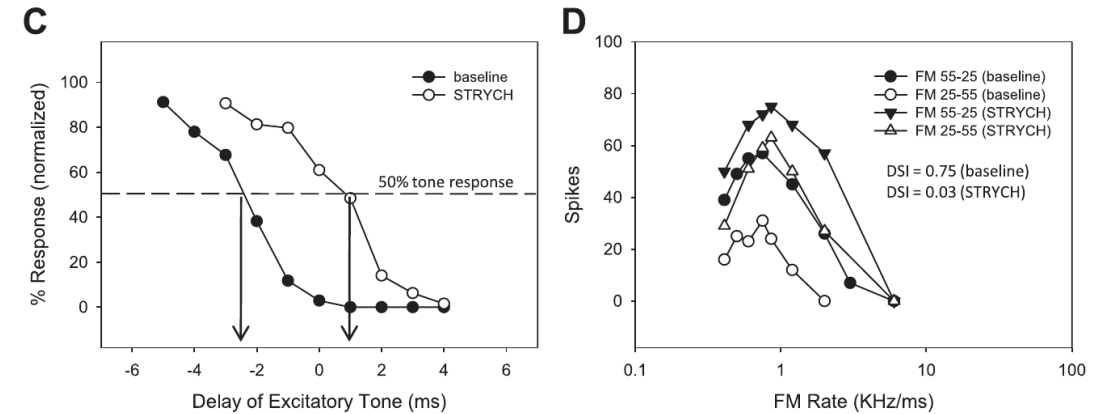
Williams & Fuzessery (2011), Fig. 6(A-B). Gabazine removes low-frequency suppression and reduces direction selectivity.

Delayed inhibition can lose its directional effect

Inhibition need not disappear completely for a neuron's stimulus preference to change. The inhibitory effect can retain substantial strength but arrive too late to prevent excitation. Selectivity depends on timing as well as on the magnitude of suppression.

In one Williams and Fuzessery example, strychnine shifted the inferred LFI arrival from -2.5 to $+1.0$ ms relative to excitation, a 3.5-ms delay. The neuron's strong downward preference was largely lost even though measurable inhibitory suppression remained.

An upward sweep could now evoke excitation before that inhibition became effective. The result supports temporal ordering of inhibitory and excitatory inputs as a determinant of direction selectivity. It does not mean that all residual inhibition is irrelevant or that receptor blockade uniformly changes input timing in every neuron.



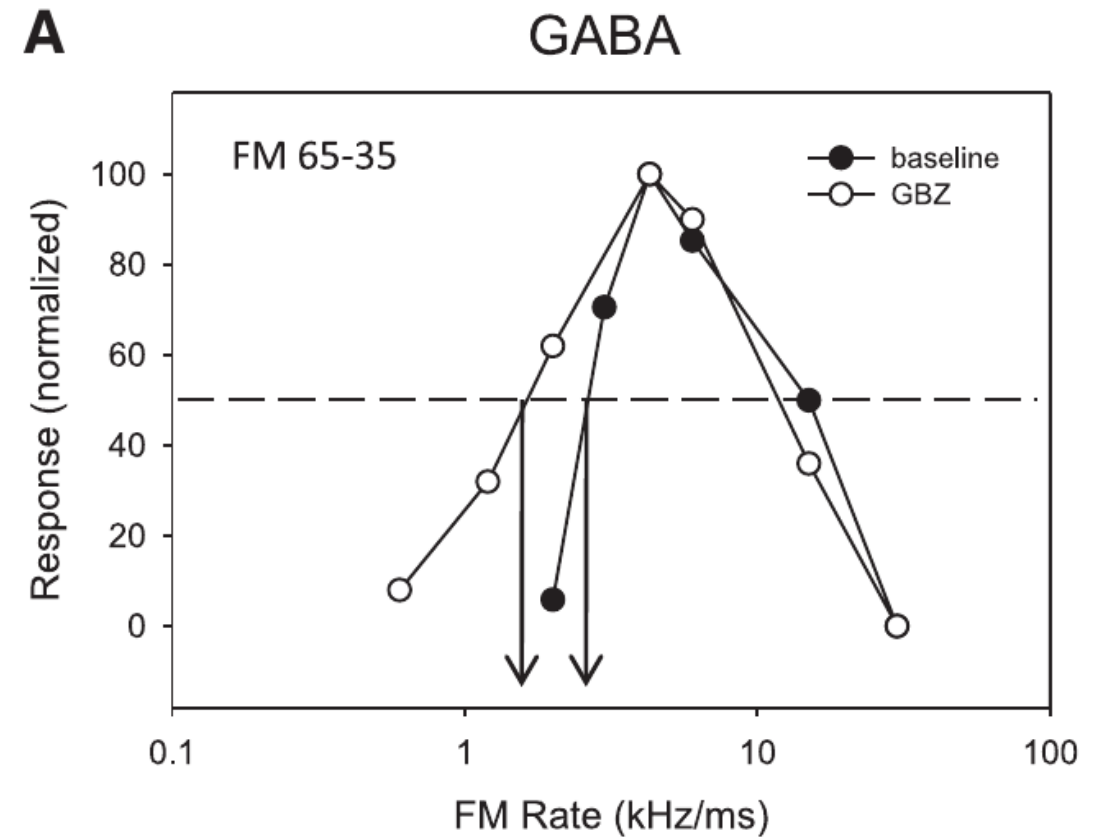
Williams & Fuzessery (2011), Fig. 6(C–D). Delayed low-frequency inhibition accompanies reduced FM direction preference.

Sweep-rate tuning is more resistant to local blockade

Fast-pass rate selectivity means that a neuron responds better to rapid FM sweeps than to sufficiently slow sweeps. Williams and Fuzessery compared this feature with direction selectivity during local inhibitory receptor blockade in the pallid-bat inferior colliculus.

Gabazine did not eliminate fast-pass selectivity in the rate-selective neurons tested, even when firing changed. Strychnine could eliminate it in a minority of neurons, allowing responses to slow downward sweeps. Direction selectivity was generally more vulnerable to either blocker.

The different outcomes indicate that FM direction and rate are not interchangeable forms of tuning. Retention of rate selectivity is consistent with inheritance from lower auditory levels in many neurons. It does not prove that local inhibitory circuitry has no influence on their responses or on their exact rate threshold.



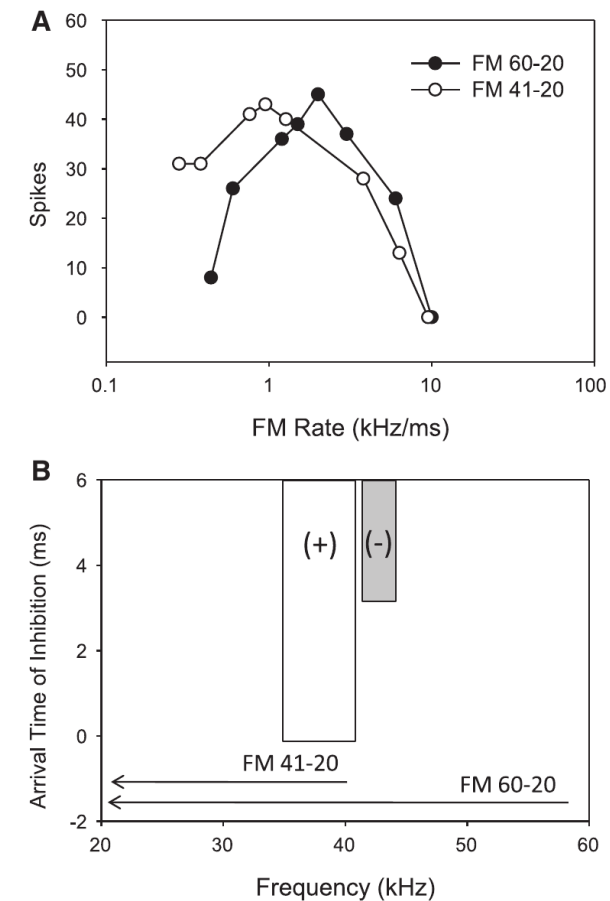
Williams & Fuzessery (2011), Fig. 7(A). Fast-pass sweep-rate preference remains during GABA_A receptor blockade.

High-frequency inhibition can reject slow sweeps

A downward sweep activates high frequencies before reaching lower excitatory frequencies. Delayed HFI can arrive after excitation during a fast sweep, allowing spikes, but before excitation during a slow sweep, preventing the response. This ordering creates a fast-pass filter for sweep rate.

Williams and Fuzessery removed the high-frequency portion of a downward sweep to test this mechanism. A neuron selective for fast sweeps began responding to slow sweeps when frequencies above 41 kHz were excluded. Those frequencies had activated its inhibitory sideband.

The response depends on both the sweep's spectral path and its duration. The experiment supports a role for HFI in rate selectivity rather than treating slow sweeps as inherently ineffective excitatory sounds. It does not identify the anatomical source of the inhibitory sideband.



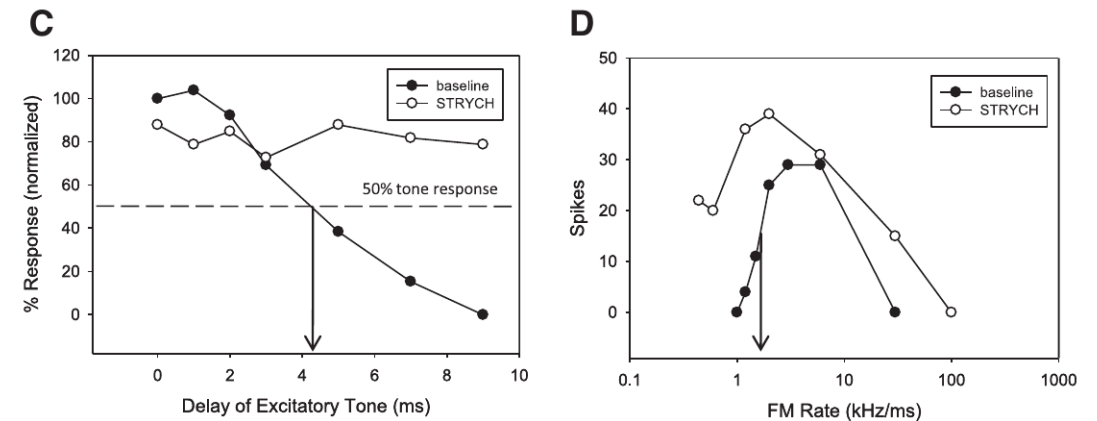
Williams & Fuzessery (2011), Fig. 8(A-B). Removing high-frequency inhibitory input permits responses to slow downward sweeps.

Glycine blockade can remove high-frequency suppression

Williams and Fuzessery paired measurements of high-frequency two-tone suppression with measurements of downward-sweep rate tuning. This tested whether the neurons losing fast-pass selectivity also lost the inhibitory sideband proposed to create it.

In a published strychnine example, HFI was eliminated and the neuron gained responses to slow sweeps. Gabazine examples retained both HFI and fast-pass selectivity. Local glycinergic input therefore supported this form of rate tuning in the affected neurons.

The finding is a selective contribution of a receptor pathway, not a universal assignment of every FM property to glycine. Many rate-selective neurons retained their preference after blockade, and the study's inference concerns sideband-based selectivity. Other rate mechanisms and upstream contributions were not eliminated by this experiment.



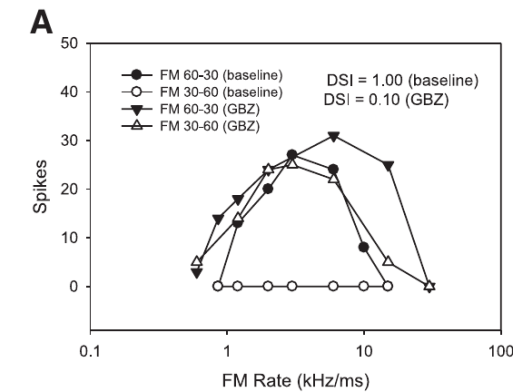
Williams & Fuzessery (2011), Fig. 9(C–D). Strychnine removes high-frequency suppression and fast-pass rate selectivity.

Local and inherited selectivity can coexist

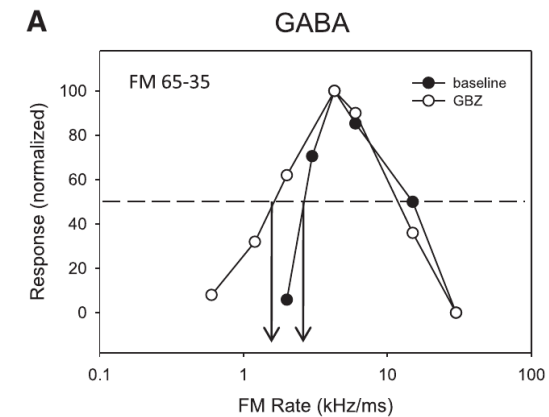
The inferior colliculus receives inhibitory and excitatory inputs from several ascending pathways as well as local and commissural connections. The paper identifies GABAergic sources including the dorsal nucleus of the lateral lemniscus; lower brainstem regions can supply both GABAergic and glycinergic inputs.

Receptor blockade near a recorded neuron tests the contribution of inhibition acting at that site. A direction preference that disappears was created or refined locally. A rate preference that persists may already be present in incoming activity, although resistant local mechanisms cannot be excluded completely.

Williams and Fuzessery inferred substantial local construction of direction selectivity and greater inheritance of sideband-based rate selectivity. They did not directly map the full presynaptic circuit. The same response property can be inherited, transformed, or reconstructed at successive auditory levels.



Williams & Fuzessery (2011), Fig. 3(A). Direction tuning before and during GABA receptor blockade.



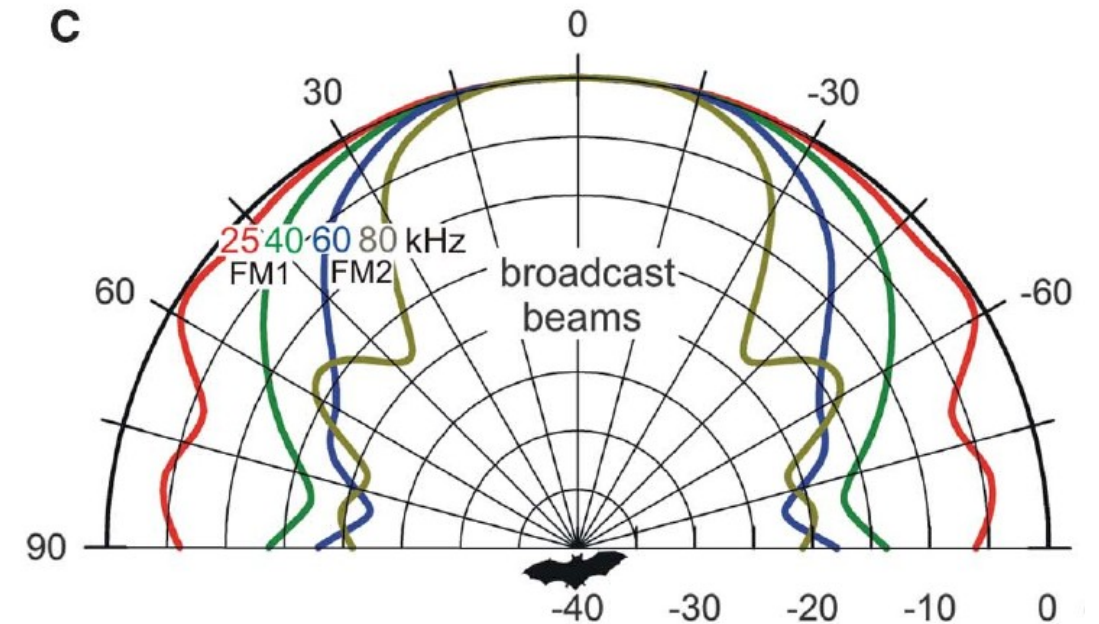
Williams & Fuzessery (2011), Fig. 7(A). FM rate selectivity persists during inhibitory receptor blockade.

The two harmonics have different beam widths

The **sonar beam** is the directional distribution of a bat's emitted sound. Its width depends on frequency. Bates and colleagues describe a horizontal beam spanning about 70° to each side at 25 kHz, compared with about 30° at 80 kHz.

A target near the beam axis receives both FM1 and FM2 strongly. Objects off to the side receive proportionally less high-frequency FM2 because its beam is narrower. The returning echo's harmonic balance therefore contains information related to where the reflecting object was ensonified.

This acoustic relationship can help distinguish a forward target from lateral **clutter**, reflections from objects irrelevant to the current target. Beam width alone does not establish successful neural segregation; the behavioral experiments tested whether bats use the resulting harmonic differences to reduce interference.



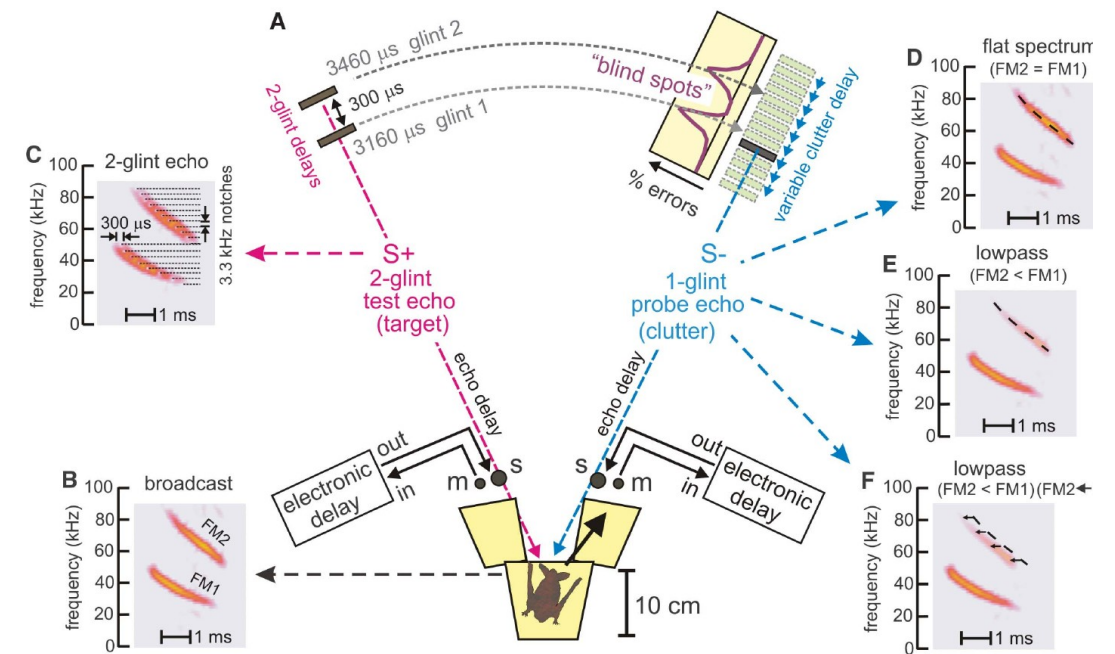
Bates et al. (2011), Fig. 1(C). Published frequency-dependent horizontal beam patterns.

A two-glinton echo provides a structured target

An **acoustic glinton** is a reflection from one part of a target. Bates and colleagues used a rewarded virtual target containing two glintons at delays of 3,160 and 3,460 μs . The 300- μs separation created a structured echo rather than one isolated reflection.

A bat on a Y-shaped platform emitted calls that microphones captured and returned through electronic delay channels. The unrewarded probe contained one glinton and was presented from the opposite loudspeaker. Its delay was varied around the two rewarded glinton delays.

The design measured how another echo interfered with perception of the target's structure and timing. The electronic target provided controlled harmonic content and delay; it was not a natural insect recording. Its value was the ability to change one echo property while retaining the rest of the task.



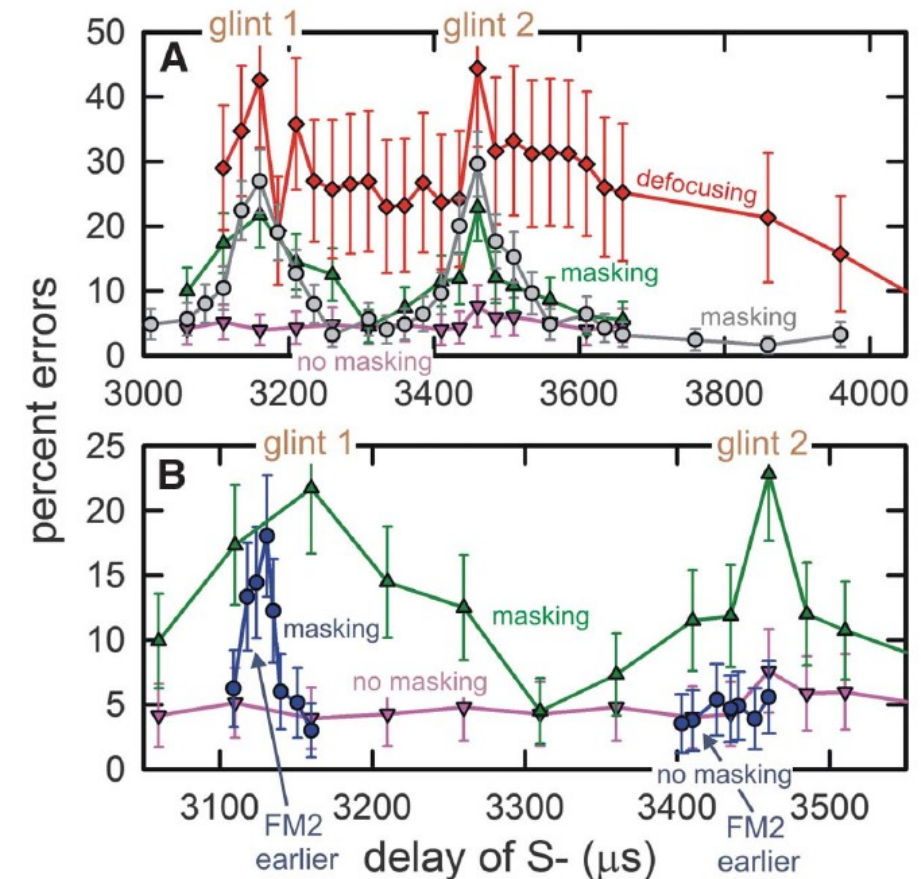
Bates et al. (2011), Fig. 2(A–F). Two-choice apparatus and controlled target and probe echoes.

Similar echo delays can create masking blind spots

Masking is interference by one sound with perception of another. In the Bates experiment, a flat-spectrum probe echo produced a sharp increase in errors when its delay approached either glint of the rewarded target. **Flat-spectrum** here means relatively balanced FM1 and FM2 content.

The masking regions extended roughly 50 μ s to either side of each target glint. A separate error peak occurred near 3,160 μ s and near 3,460 μ s. Baseline errors were low when the probe's delay lay outside these regions.

Temporal proximity can therefore disrupt a target judgment even when echoes come from different directions. The operational “blind spot” is a delay region with impaired discrimination, not anatomical damage or complete inability to hear sound. The experiment measured the chosen target's perceptual interference under controlled feedback conditions.



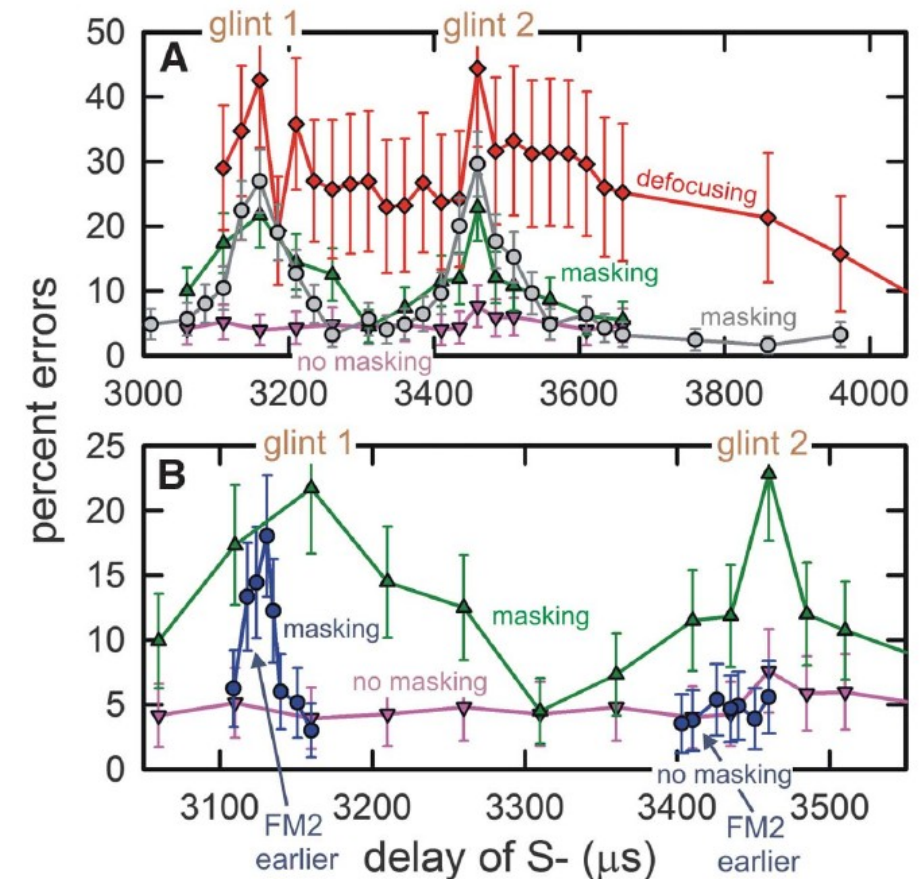
Bates et al. (2011), Fig. 3(A–B). Discrimination errors peak near the two target-glint delays.

Weakening FM2 can prevent a probe from masking

Low-pass filtering weakens higher frequencies more than lower frequencies. Bates and colleagues applied mild low-pass filtering to the probe echo, reducing FM2 relative to FM1 by an average of about 1.5 dB. This approximated harmonic imbalance in nearby lateral clutter.

The filtered probe no longer produced the masking peaks caused by the balanced probe at similar delays. The experiment retained the rewarded target's structure and changed the interfering echo, so the improvement was not caused by making the target itself easier through a new target delay.

A small harmonic imbalance could therefore strongly alter interference in the task. The result supports use of echo spectral structure for clutter rejection. It does not imply that every natural clutter echo has the same spectrum or that filtering always leaves target information intact.



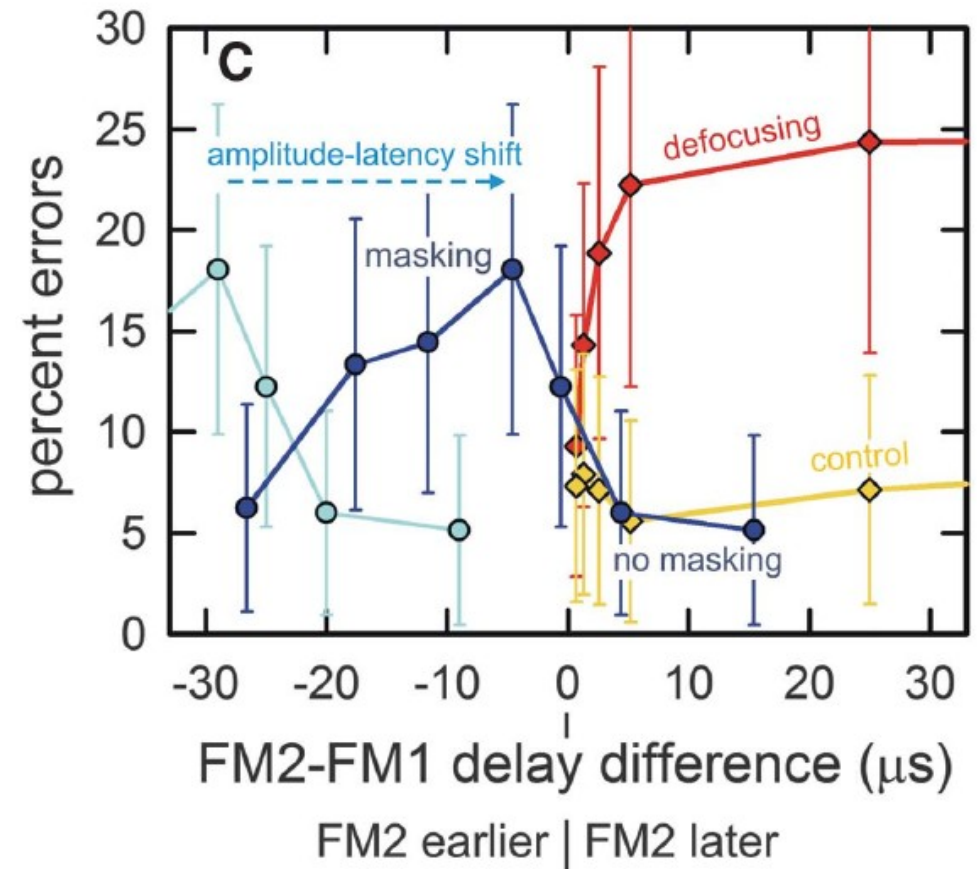
Bates et al. (2011), Fig. 3(A–B). Low-pass probe echoes reduce masking; timing compensation can restore it.

Harmonic timing links acuity and clutter rejection

Bates and colleagues proposed that weaker FM2 evokes later neural responses through amplitude–latency trading. Responses to the two harmonics then become misaligned. They interpreted this mismatch as **defocusing**, loss of a sharply specified echo-delay percept.

For the roughly 1.5-dB FM2 reduction, the proposed relative latency shift was about 24 μs . Advancing FM2 electronically by approximately 25–45 μs restored masking of the first target glint, consistent with compensation for that level-dependent timing shift.

The manipulation links harmonic timing to the trade-off between precise delay perception and resistance to clutter. Neural responses were inferred from prior timing measurements rather than recorded in this behavioral experiment. Recovery differed between the two target glints, so the result is not a uniform restoration of every part of the target image.



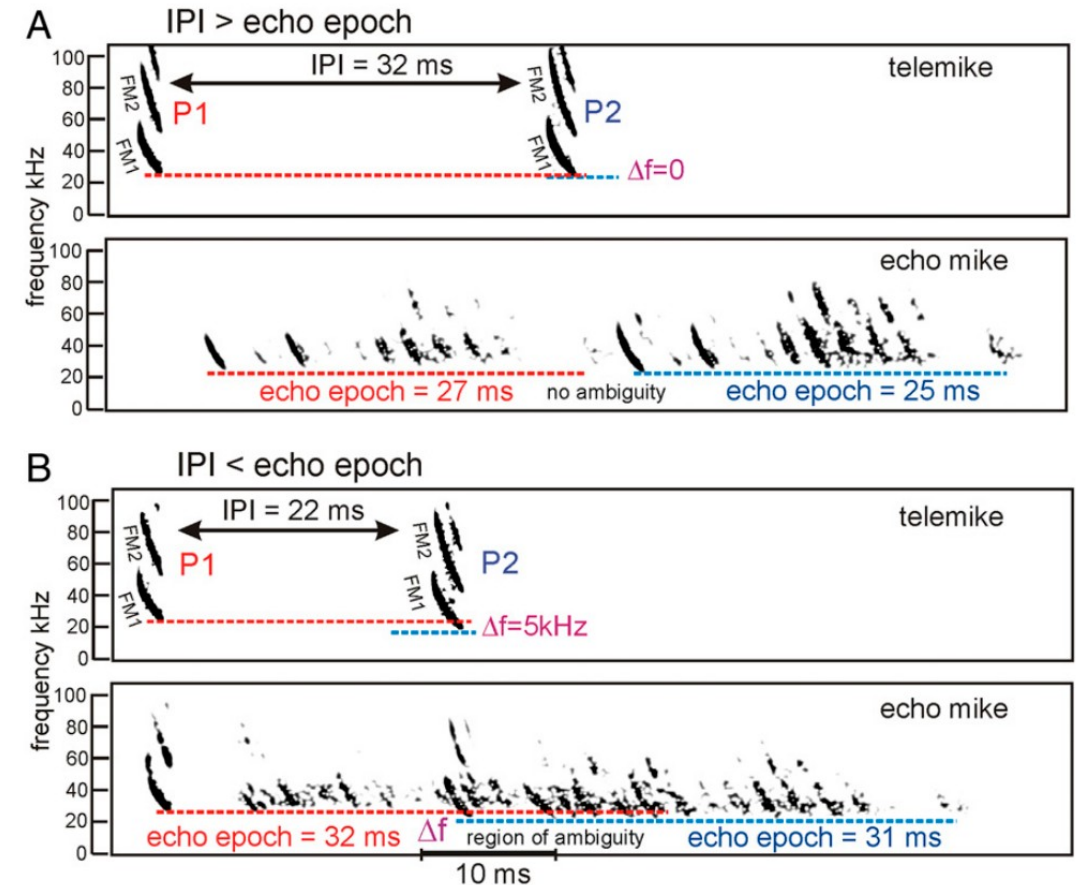
Bates et al. (2011), Fig. 3(C). Harmonic timing differences relate delay defocusing to masking recovery.

Rapid calling can mix echoes from different broadcasts

The **interpulse interval**, or IPI, is the time between successive broadcasts. The **echo epoch** is the period during which reflections from one broadcast return from objects distributed across a scene. A long-range reflector may return sound after a nearer one.

Ming and colleagues discuss cases in which the IPI becomes shorter than the echo epoch. Their published examples compare a 32-ms IPI with shorter echo epochs and a 22-ms IPI with overlapping epochs. Echoes from one broadcast can arrive after the next broadcast has occurred.

This creates **pulse–echo ambiguity**, uncertainty about which broadcast generated a reflection. Measuring delay from the wrong broadcast assigns an incorrect range. Rapid updates needed for nearby obstacles can therefore conflict with waiting for all distant echoes to finish returning.



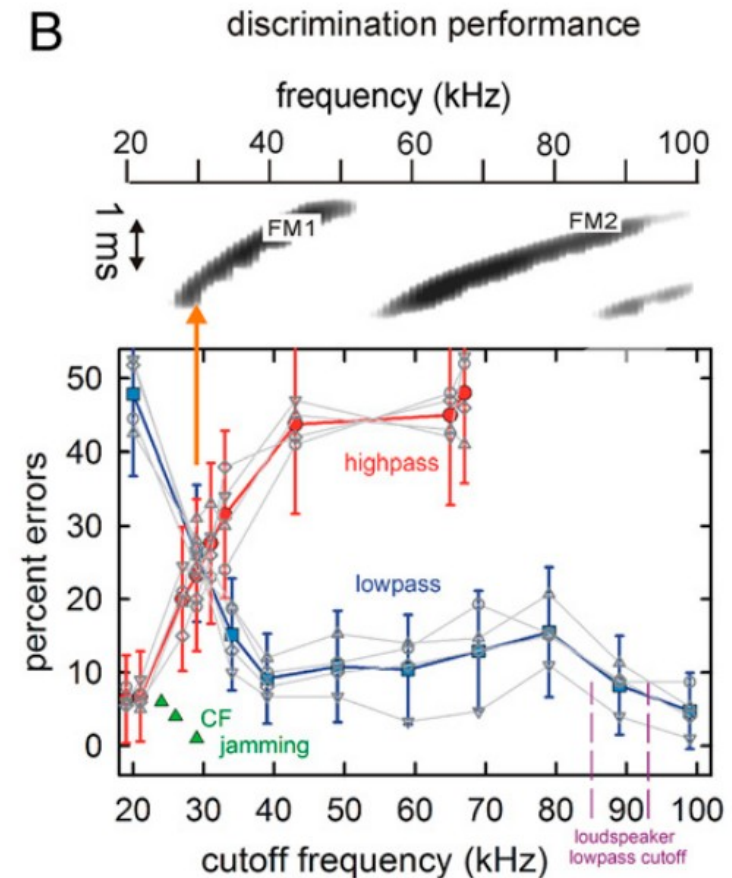
Ming et al. (2020), Fig. 2(A–B). Separate and overlapping echo epochs at different interpulse intervals.

Lowest FM1 frequencies anchor the tested delay percept

Ming and colleagues used a two-choice virtual-echo task to test which frequency regions supported target discrimination. They progressively removed either high or low frequencies from the rewarded two-glitch echo while keeping the unrewarded comparison echo broadband.

Low-pass filtering left discrimination effective when only approximately 25–30 kHz remained. High-pass filtering that removed this low-frequency end disrupted performance even while most higher FM1 frequencies and FM2 remained.

The authors proposed that **frequency hopping**, shifts of several kilohertz between broadcasts, could use this low-frequency anchor to segregate overlapping echo streams. The filtering result is measured behavior; the full pulse-assignment mechanism and its SCAT receiver implementation remain proposed or modeled explanations rather than direct neural recordings.



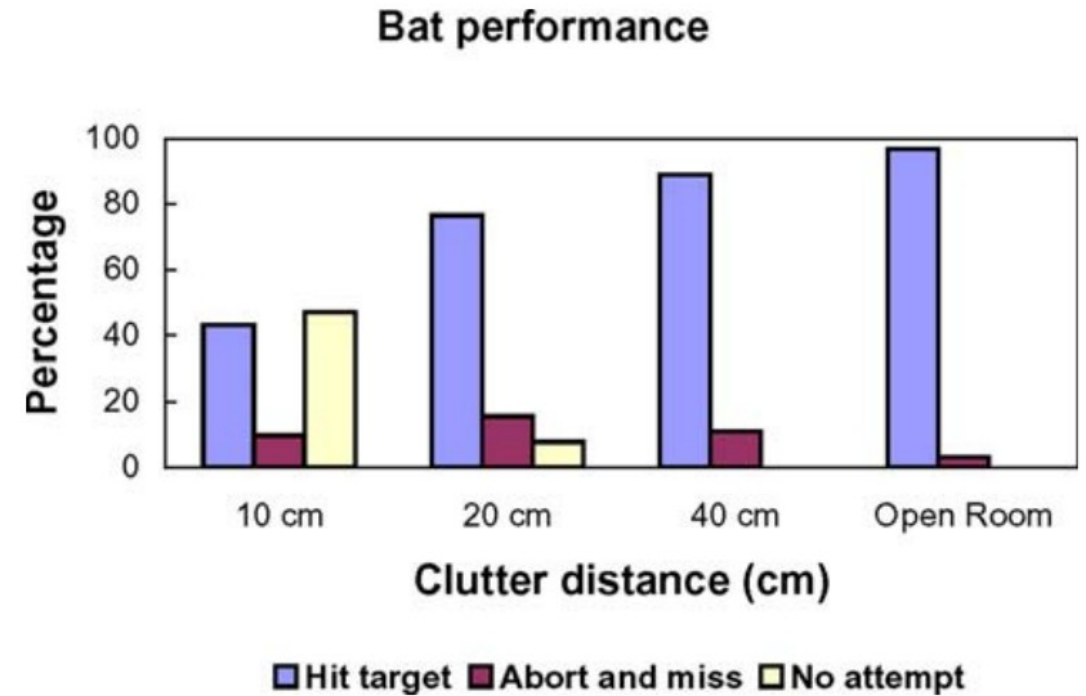
Ming et al. (2020), Fig. 3(B). Opposite effects of low-pass and high-pass filtering on echo discrimination.

Nearby vegetation reduces prey-capture success

Moss and colleagues studied freely flying big brown bats capturing tethered insects near a plant. Synchronized ultrasound recordings and high-speed video linked emitted sounds to reconstructed three-dimensional flight paths. The open-room condition provided a comparison without the plant's nearby echo clutter.

With the insect 10 cm from vegetation, bats hit or captured it in about 40% of trials and made no attempt in nearly half. Hits or captures increased to almost 80% at 20 cm and about 90% at 40 cm. Open-room performance was nearly complete.

Clutter therefore affected both willingness to attempt interception and success after approach. The combined hit/capture measure is broader than successful consumption. The experiment reveals difficulty under controlled vegetation spacing rather than a universal minimum separation for catching prey in natural habitats.



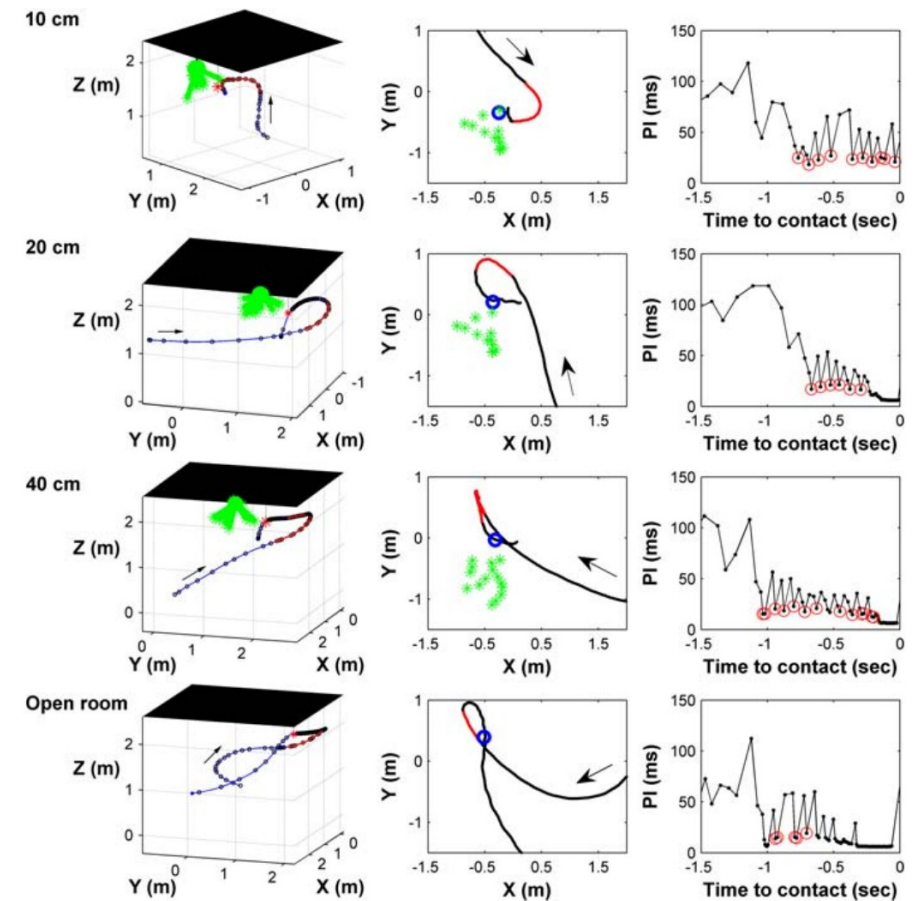
Moss et al. (2006), Fig. 2. Interception outcomes at three plant–target separations and in an open room.

Flight paths change target–clutter separation

Bats in the Moss experiment often flew alongside the vegetation and insect before turning toward the target. The approach was not simply a straight line through the plant's strongest echo direction. Flight geometry altered the angular relationship between prey and background.

As the target was placed closer to vegetation, the final direct approach became shorter and the late turn more abrupt. Three-dimensional trajectories and emitted pulse intervals were measured together. Strong changes in turning occurred mainly during the final 50–100 ms before contact.

The authors interpreted these trajectories as reducing interference by ensonifying prey off axis from vegetation during part of the approach. The paths are measured behavior, whereas that masking-reduction function is an interpretation supported by geometry. It was not tested by recording the bat's full internal spatial representation.



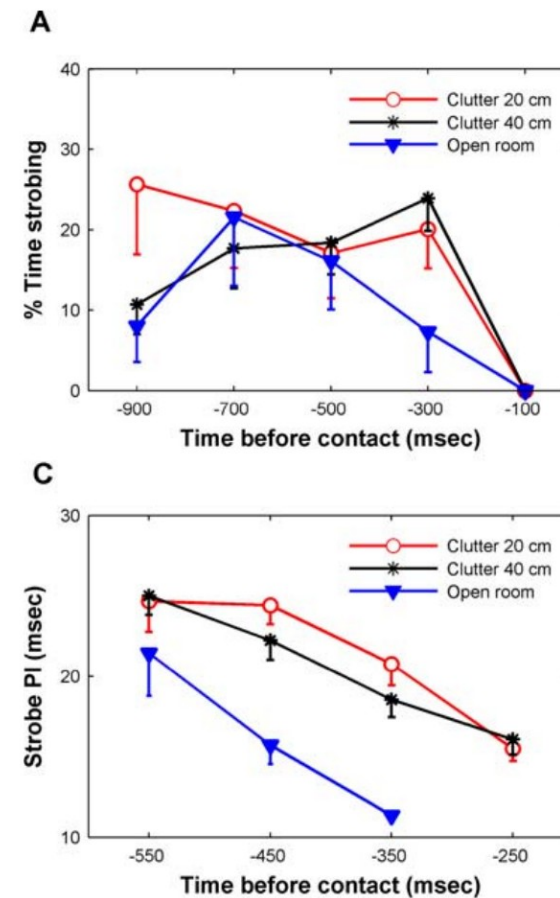
Moss et al. (2006), Fig. 4. Measured flight paths and pulse intervals for clutter and open-room trials.

Sonar strobe groups control repeated scene sampling

Sonar strobe groups are clusters of calls separated from neighboring calls or groups by longer intervals. Moss and colleagues identified groups with stable internal intervals when they contained three or more sounds. This temporal organization occurred during approach before the terminal capture phase.

Within groups, pulse intervals shortened as the bat approached its target. Calls in the clutter conditions tended to be grouped at shorter intervals than calls in the open room. The group pattern added structure beyond an overall increase in call rate.

The authors proposed that repeated sampling at stable intervals could sharpen the spatial representation of the scene. The study measured vocal timing and behavior, not the proposed improvement in a neuronal population code. Grouping is therefore an established motor pattern with a hypothesized perceptual function.



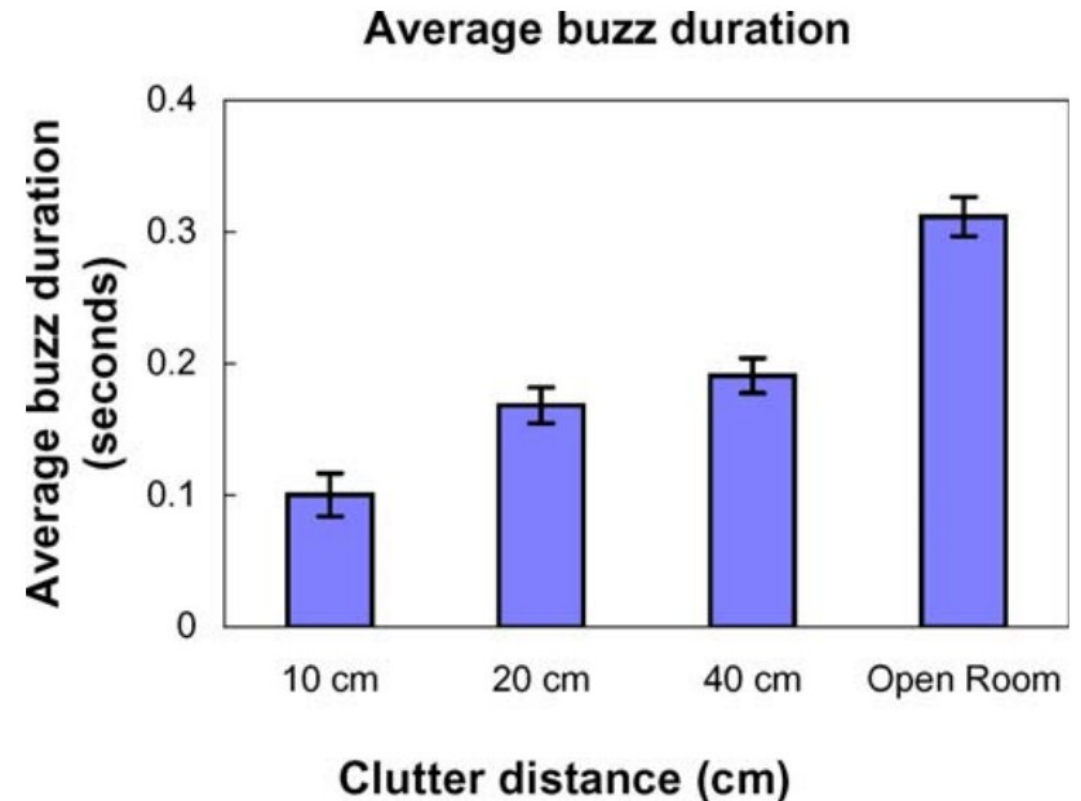
Moss et al. (2006), Fig. 10(A,C). Strobing occurrence and within-group pulse intervals during approach.

The terminal buzz shortens near vegetation

The **terminal buzz** is the rapid-call sequence immediately before prey interception. Moss and colleagues defined buzz-II calls by pulse intervals of 10 ms or less and compared its duration across plant–target separations and the open-room condition.

Mean buzz-II duration declined from over 300 ms in the open room to about 200 ms at 40 cm from vegetation and about 100 ms at 10 cm. Prey capture was still sometimes successful with the shorter buzz, including at the closest tested separation.

The authors proposed that extended clutter echoes reduce the usefulness of prolonged rapid calling. The observed adjustment connects vocal timing to scene structure, but it does not directly establish the neural masking mechanism. Successful interception can involve coordinated changes in call groups, final approach geometry, and buzz duration.



Moss et al. (2006), Fig. 8. Buzz-II duration varies with target–vegetation separation.

Key takeaways

Echo delay carries target range. Controlled electronic-delay experiments support relative-range judgments without changing the reflector's physical distance.

Timing precision and latency differ. Population responses can arrive after milliseconds while repeating within tens of microseconds; weaker sounds can shift their latency.

Inhibition selects FM features. Early low-frequency inhibition shapes direction preference; delayed high-frequency inhibition can reject slow downward sweeps.

Harmonic structure affects clutter interference. Weakening FM2 can reduce masking, and compensating harmonic timing can restore part of that interference.

Call assignment is part of ranging. Overlapping echo epochs create pulse–echo ambiguity; lowest FM1 frequencies anchor the tested delay percept.

Active sensing changes the scene. Bats adjust flight paths, call groups, and terminal-buzz duration near vegetation; proposed neural benefits remain distinct from measured behavior.